

Fundamentals of Solid State Physics

Thermal Properties

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Previous Classes

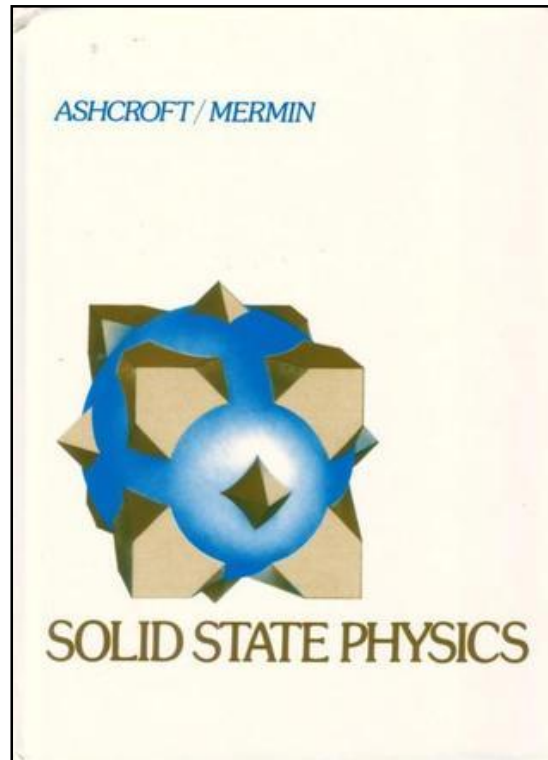
- Introduction (Week 1)
- Materials and Crystal Structures (Week 2–3)
- **Electronic Properties (Week 4–12)**
 - Free electrons (the Drude and Sommerfeld models)
 - Electrons in a periodic potential (Bloch's Theorem)
 - The near-free electron model, the tight-binding model
 - Electronic band diagram, band gaps, effective mass
 - Metals, insulators, semiconductors
 - Devices: junctions, diodes, transistors
- Thermal Properties (Week 13)
- Optical Properties (Week 14)
- Magnetic Properties (Week 15)

This Class

- Introduction (Week 1)
- Materials and Crystal Structures (Week 2–3)
- Electronic Properties (Week 4–12)
- **Thermal Properties (Week 13)**
 - ▣ Crystal vibration, phonon band
 - ▣ Thermal capacity and conductivity
- Optical Properties (Week 14)
- Magnetic Properties (Week 15)

Further Reading

- **Ashcroft & Mermin, Chapter 21, 22, 23**



Born-Oppenheimer Approximation

- Adiabatic Approximation 绝热近似
- Static Approximation 定核近似
 - The behaviors of electrons and nuclei can be calculated separately.

$$\Psi_{\text{total}} = \Psi_{\text{electron}} * \Psi_{\text{nuclear}}$$

- Electrons move much faster than nuclei
- When we consider the electronic behaviors, we assume the atomic lattice is static.



Failures of the Static Lattice Model

It cannot explain

- **Scattering of electrons**
 - temperature dependence of σ
- **Thermal properties**
 - Thermal Capacity
 - Thermal Conductivity
 - Thermal Expansion
- **Mechanical properties**
- ...

$$\sigma = ne\mu = ne^2 \frac{\tau}{m}$$

τ - relaxation time (s)

***We have to analyze
lattice vibration***



How to analyze Lattice Vibration?

- Born-Oppenheimer Approximation
- The behaviors of electrons and nuclei can be calculated separately.

$$\Psi_{\text{total}} = \Psi_{\text{electron}} * \Psi_{\text{nuclear}}$$

- When we consider the electronic behaviors, we assume the atomic lattice is static.
- When we consider the lattice behaviors, we assume electrons are static.

Fundamentals of Solid State Physics

Lattice Vibration - Classical Model

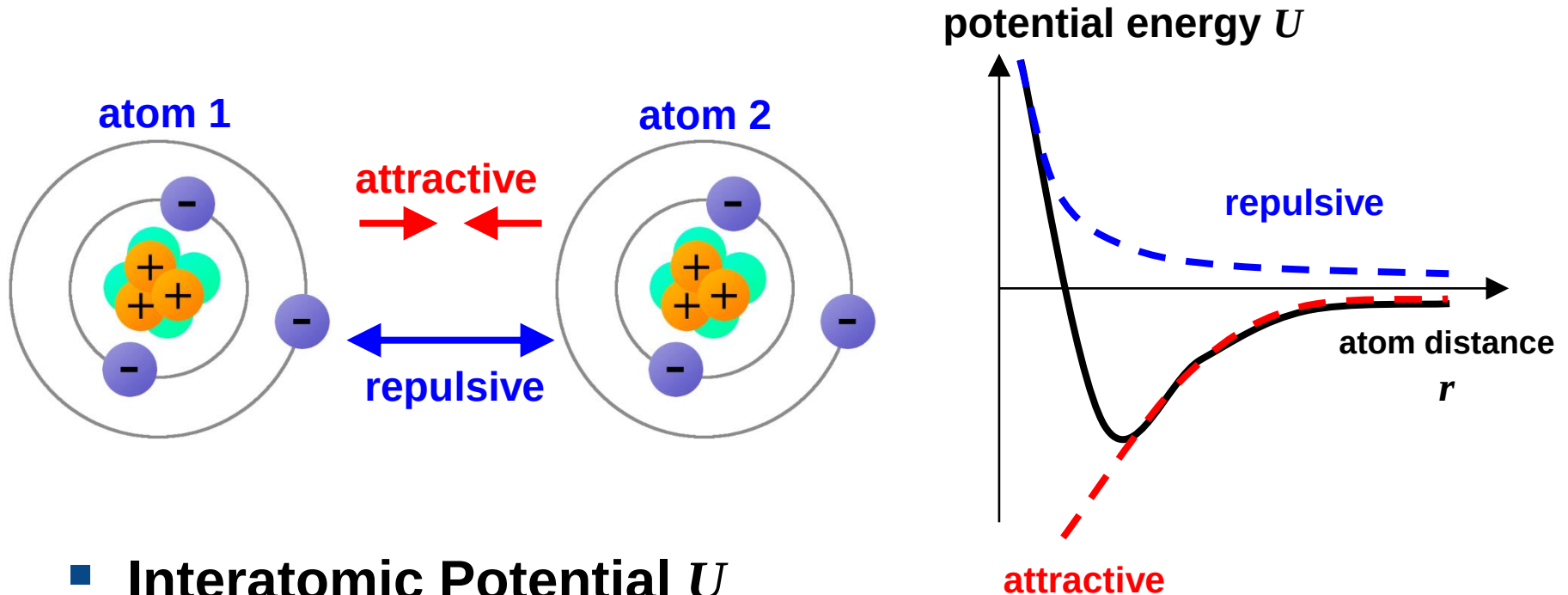
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Atomic Interactions



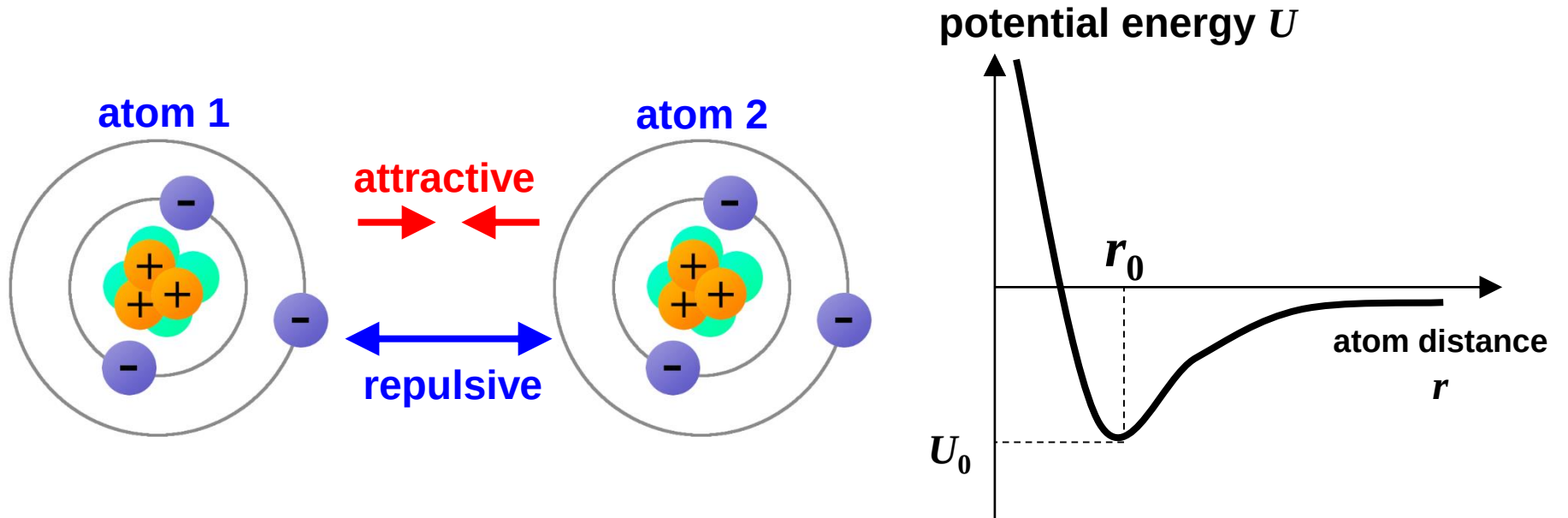
■ Interatomic Potential U

$$U(r) = -U_{\text{attraction}}(r) + U_{\text{repulsion}}(r)$$

U - potential energy (J, eV)

r - atomic distance (nm, Å)

Atomic Interactions



■ Interatomic Potential U

$-U_0$ - cohesive energy (结合能)
lowest energy state

$$U(r) = -U_{\text{attraction}}(r) + U_{\text{repulsion}}(r)$$

U - potential energy (J, eV)

r - atomic distance (nm, Å) 12

Examples

■ Lennard-Jones (L-J)

$$U(r) = -\frac{A}{r^6} + \frac{B}{r^{12}}$$

attractive

repulsive

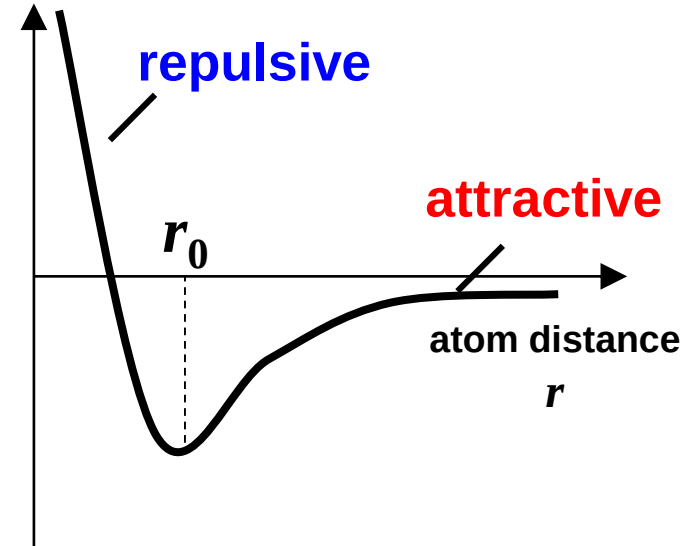
■ Buckingham Potential

$$U(r) = -\frac{A}{r^6} + B \exp\left(-\frac{r}{\rho}\right)$$

attractive

repulsive

potential energy U



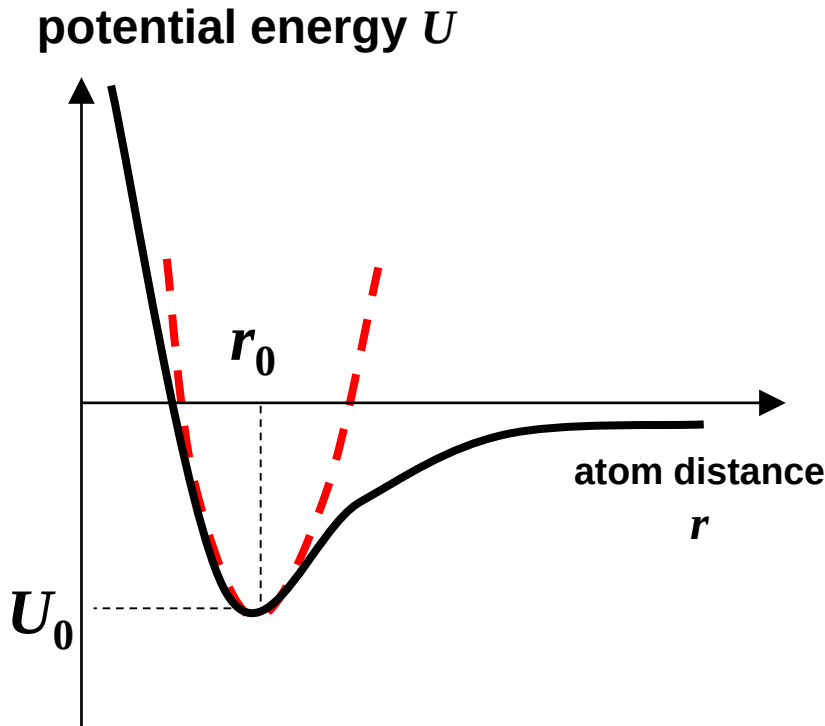
■ Morse Potential

$$U(r) = D \left(-2e^{-a(r-r_0)} + e^{-2a(r-r_0)} \right)$$

attractive

repulsive

Atomic Interactions



*Harmonic Approximation
(parabolic curve)*

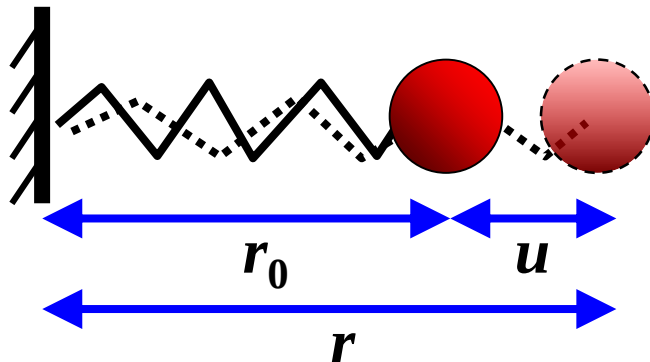
$$U(r) = U_0 + \frac{1}{2} K (r - r_0)^2$$

$$= U_0 + \frac{1}{2} K \cdot u^2$$



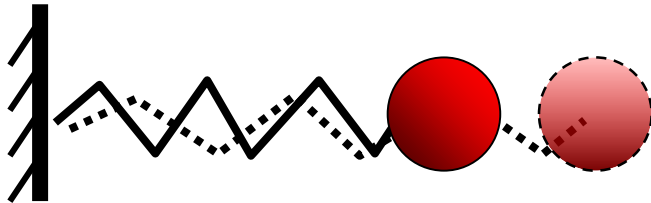
$$F = -\frac{\partial U}{\partial r} = -K \cdot u$$

Hooke's Law



K - spring constant (N/m)
 u - atomic displacement (m) 14

Harmonic Oscillator 谐振子



$$F = -\frac{\partial V}{\partial r} = -K \cdot u$$

Hooke's Law

$$F = m \frac{\partial v}{\partial t} = m \frac{\partial^2 u}{\partial t^2}$$

Newton's Second Law



$$m \frac{\partial^2 u}{\partial t^2} + Ku = 0$$

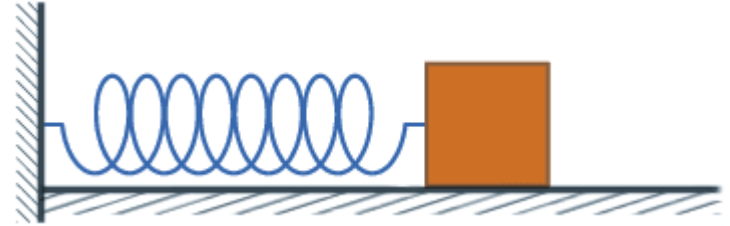
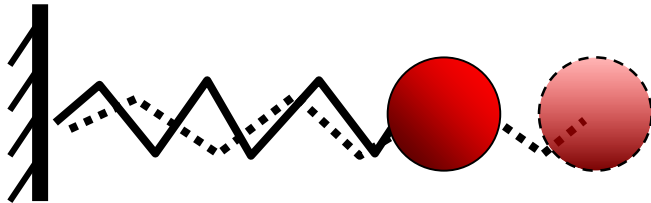


$$u = Ae^{-i\omega t}$$

$$\omega = \sqrt{\frac{K}{m}}$$

ω - angular frequency (rad/s) 15

Harmonic Oscillator 谐振子



$$F = -\frac{\partial V}{\partial r} = -K \cdot u$$

Hooke's Law

$$F = m \frac{\partial v}{\partial t} = m \frac{\partial^2 u}{\partial t^2}$$

Newton's Second Law



$$m \frac{\partial^2 u}{\partial t^2} + Ku = 0$$



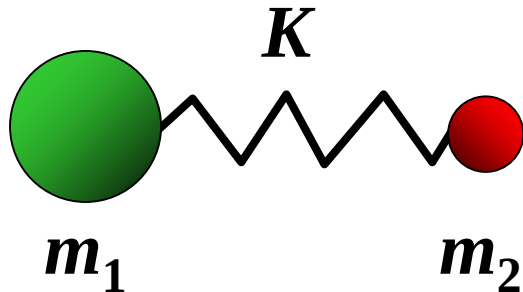
$$u = Ae^{-i\omega t}$$

$$\omega = \sqrt{\frac{K}{m}}$$

ω - angular frequency (rad/s) 16

Diatomic Molecule 双原子分子

Homework 8.1



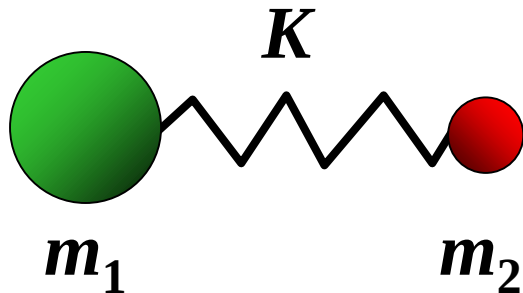
$$\omega = \sqrt{\frac{K}{m^*}} = \sqrt{\frac{m_1 + m_2}{m_1 m_2} K}$$

ω - angular frequency (rad/s)

m^* - reduced mass (kg)

Diatomic Molecule 双原子分子

Homework 8.1

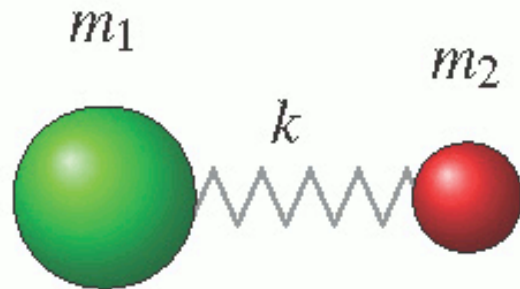


$$\omega = \sqrt{\frac{K}{m^*}} = \sqrt{\frac{m_1 + m_2}{m_1 m_2} K}$$

ω - angular frequency (rad/s)

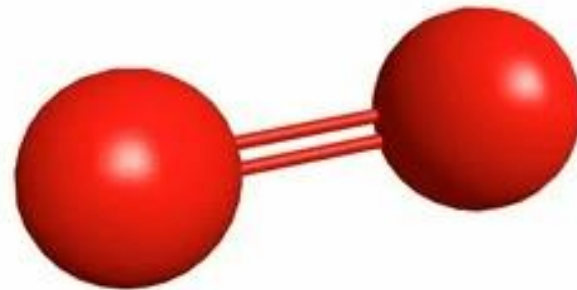
m^* - reduced mass (kg)

CO molecule



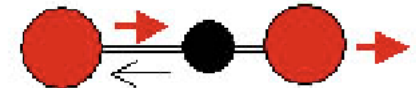
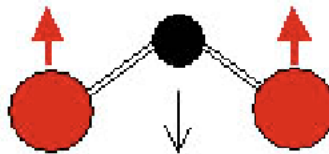
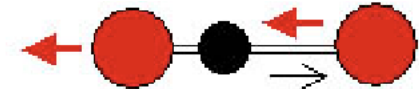
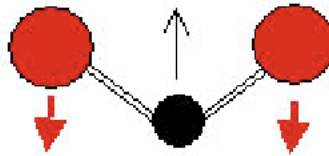
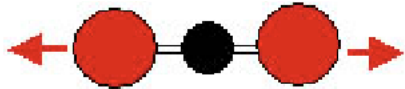
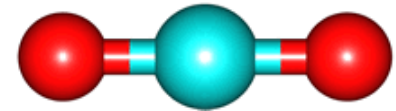
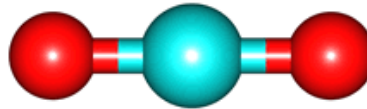
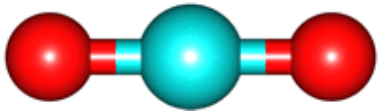
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O₂ molecule



MakeAGIF.com

CO₂ Vibration



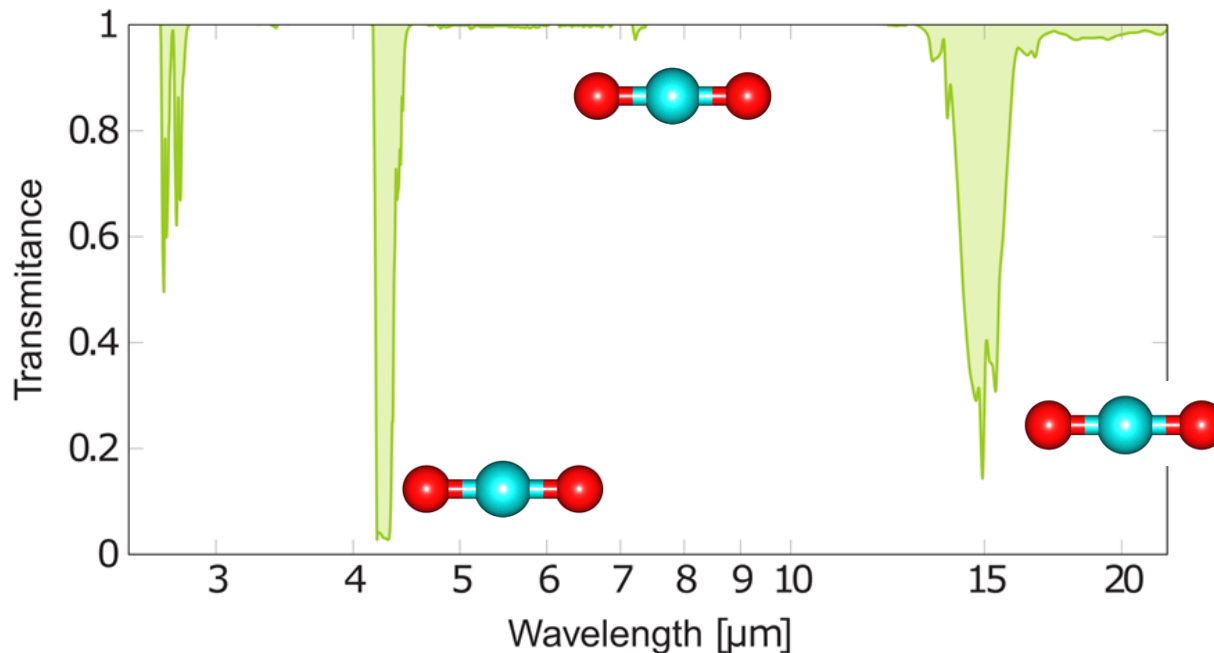
Symmetric Stretch
1366 cm⁻¹ or 7.32 μm

Bending
667 cm⁻¹ or 15 μm

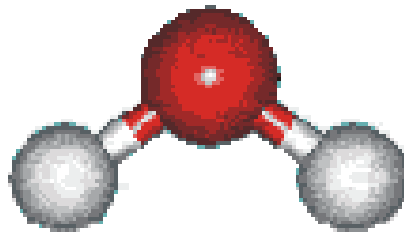
Asymmetric Stretch
2349 cm⁻¹ or 4.23 μm

Molecule Vibration

- Vibration modes of molecules can be measured by **infrared absorption spectrum**
- Infrared absorption of CO₂ causes **greenhouse effect**

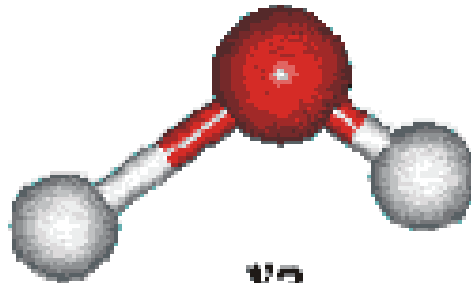


H₂O Vibration



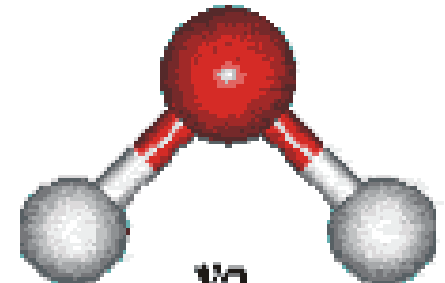
v₁

symmetric stretch



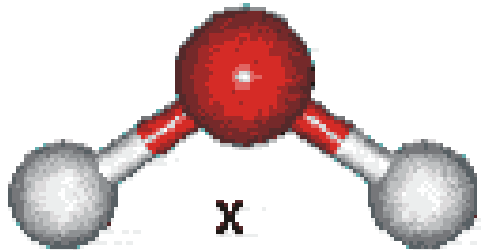
v₃

asymmetric stretch

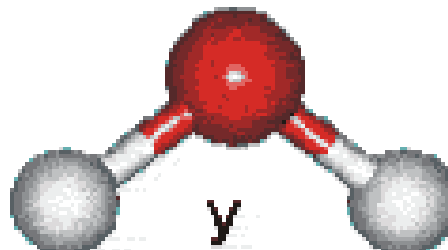


v₂

bend

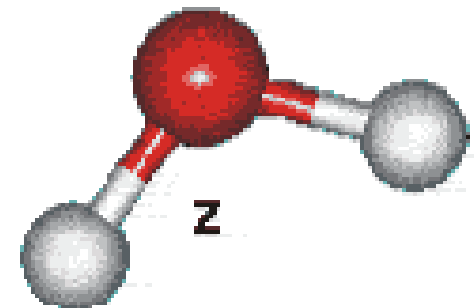


x



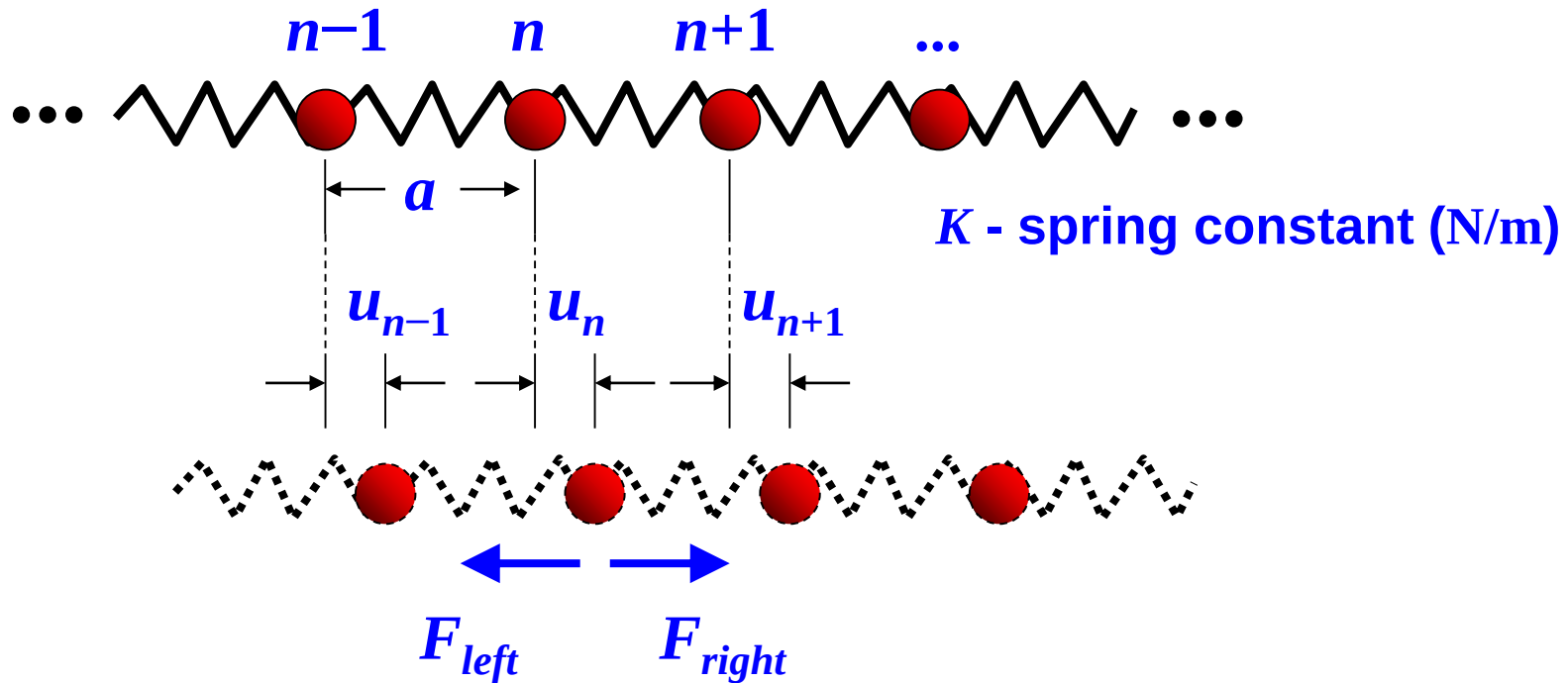
y

librations



z

1D *Monatomic* Chain 单原子链



$$F_{left} = K(u_n - u_{n-1})$$

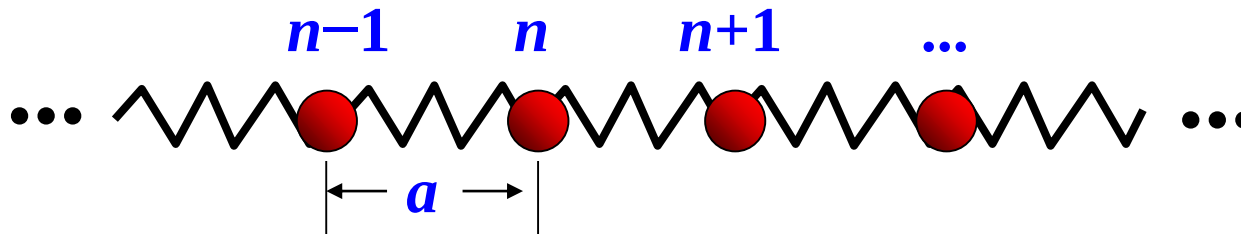
$$F_{right} = K(u_{n+1} - u_n)$$

$$F_n = F_{left} - F_{right}$$

and

$$F_n = m \frac{\partial^2 u_n}{\partial t^2}$$

1D *Monatomic* Chain 单原子链



$$m \frac{\partial^2 u_n}{\partial t^2} + K(u_n - u_{n-1}) + K(u_n - u_{n+1}) = 0$$



$$u_n = Ae^{i(kx - \omega t)}$$

$$u_{n\pm 1} = Ae^{i(kx - \omega t)} e^{\pm ika}$$

Acoustic / Sound Wave (声波)

Elastic Wave (弹性波)

Mechanical Wave (机械波)

Lattice Wave (格波)

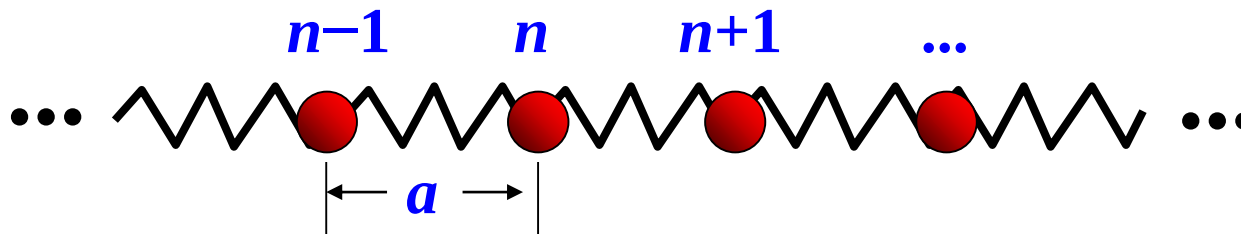
...

Wave Function

k - wave vector (m^{-1})

ω - angular frequency (Hz) 23

1D *Monatomic* Chain 单原子链

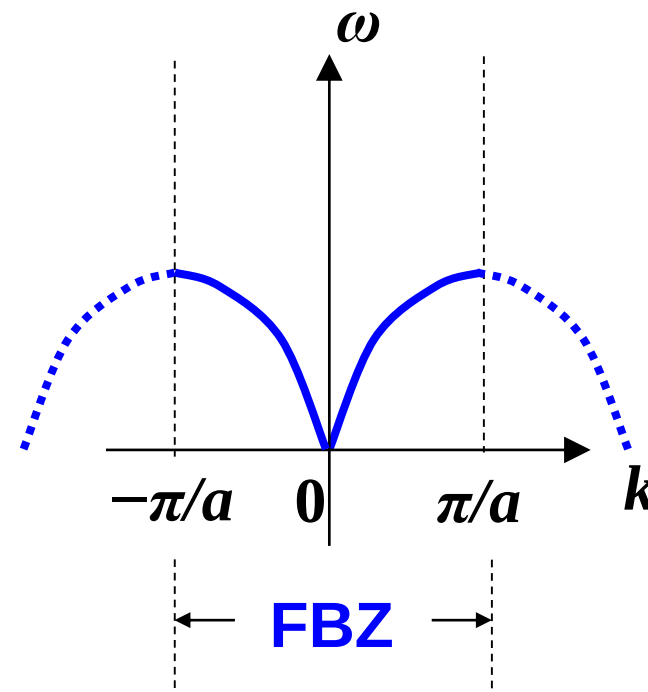


$$\begin{aligned}\omega^2 &= \frac{K}{m} (2 - e^{-ika} - e^{+ika}) \\ &= \frac{4K}{m} \sin^2\left(\frac{ak}{2}\right)\end{aligned}$$

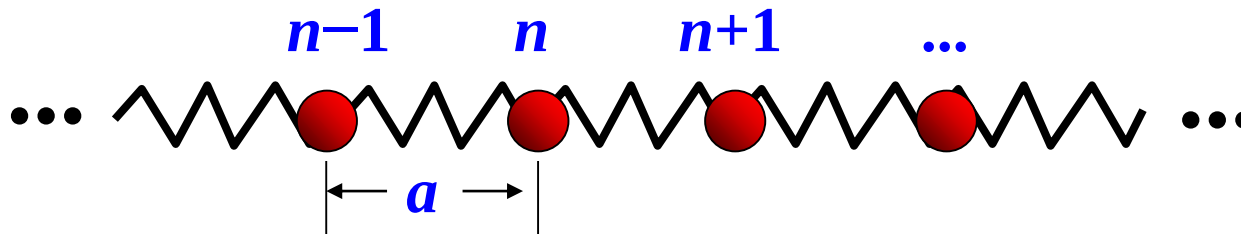
→

$$\omega = \sqrt{\frac{4K}{m}} \left| \sin\left(\frac{ak}{2}\right) \right|$$

ω - k diagram
(dispersion curve)



1D *Monatomic* Chain 单原子链



ω - k diagram
(dispersion curve)

$$k = 6\pi/6a \quad \lambda = 2.00a \quad \omega_k = 2.00\omega$$



$$k = 5\pi/6a \quad \lambda = 2.40a \quad \omega_k = 1.93\omega$$



$$k = 4\pi/6a \quad \lambda = 3.00a \quad \omega_k = 1.73\omega$$



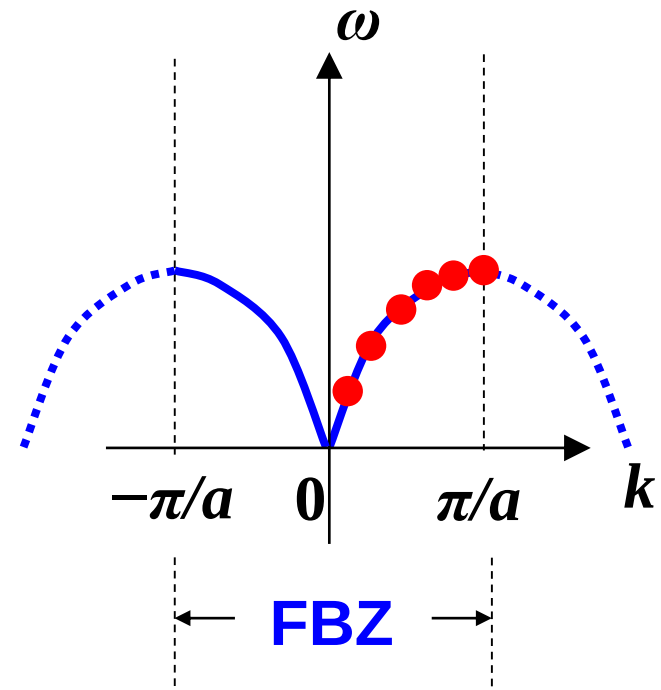
$$k = 3\pi/6a \quad \lambda = 4.00a \quad \omega_k = 1.41\omega$$



$$k = 2\pi/6a \quad \lambda = 6.00a \quad \omega_k = 1.00\omega$$



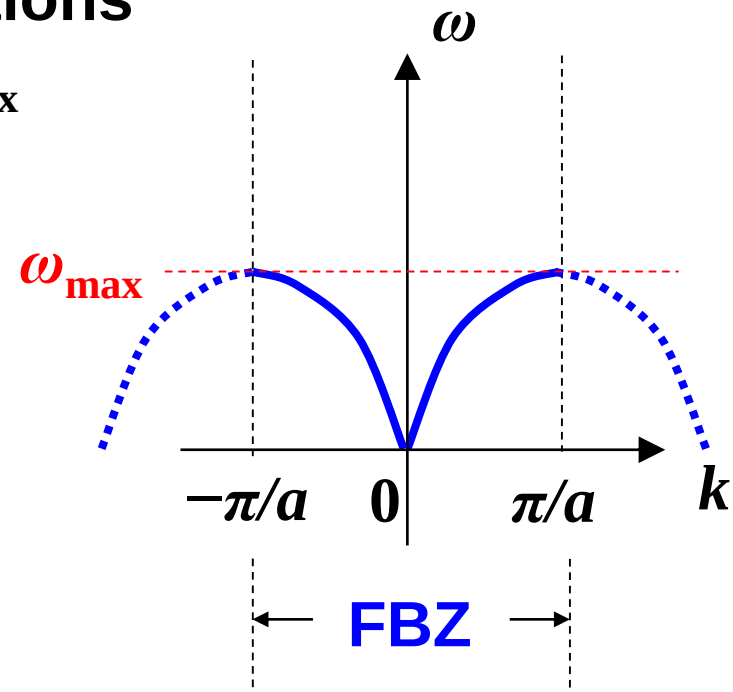
$$k = \pi/6a \quad \lambda = 12.00a \quad \omega_k = 0.52\omega$$



1D *Monatomic* Chain 单原子链

- Crystals cannot transmit vibrations above the cutoff frequency $\omega_{\max}$

$$\omega_{\max} = \sqrt{\frac{4K}{m}}$$



- Phase velocity

$$v_p = \omega / k$$

- Group velocity

$$v_g = \partial\omega / \partial k$$

- Standing wave ($v_g = 0$) at

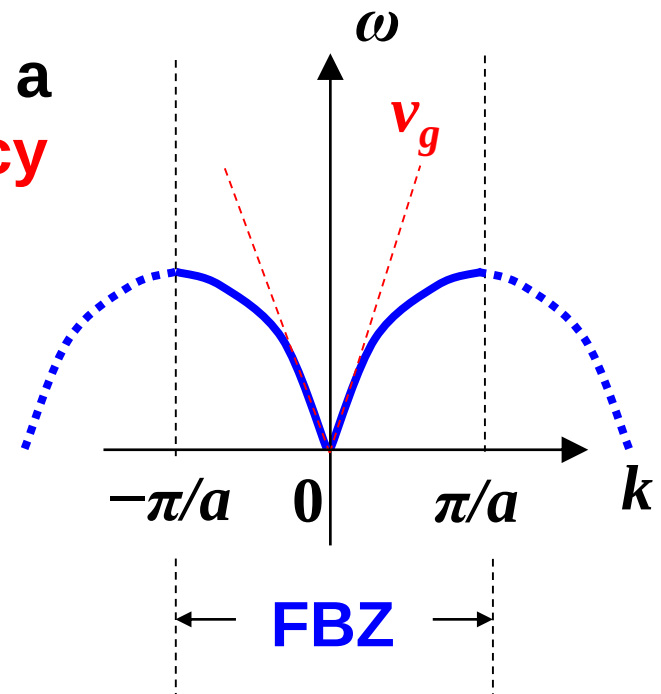
$$k = \pm \frac{\pi}{a}$$

Speed of Sound 声速

- At long wavelength limit $ka \approx 0$, group velocity is **speed of sound**, a constant **independent of frequency**

$$\omega = \sqrt{\frac{4K}{m}} \left| \sin\left(\frac{ak}{2}\right) \right| \approx \sqrt{\frac{K}{m}} ak$$

$$\begin{aligned} v_g &= \partial\omega / \partial k \\ &= a \sqrt{\frac{K}{m}} \end{aligned}$$

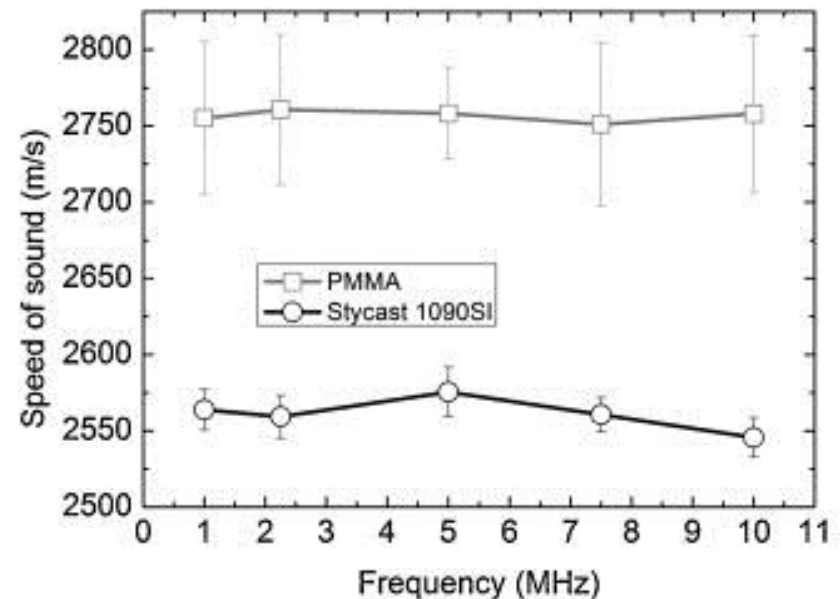


Speed of Sound 声速

- At long wavelength limit $ka \approx 0$, group velocity is **speed of sound**, a constant **independent of frequency**

sound speed in solids

$$v_g = \partial \omega / \partial k$$
$$= a \sqrt{\frac{K}{m}}$$



Speed of Sound 声速

- At long wavelength limit $ka \approx 0$, group velocity is **speed of sound**, a constant **independent of frequency**

$$v_g = \partial \omega / \partial k$$

$$= a \sqrt{\frac{K}{m}}$$

- Higher v_g requires
 - stronger bonds (K)
 - smaller atoms (m)

Material	Speed of Sound (m/s)
Rubber	60
Pb	1210
Au	3240
Cu	4600
Al	6320

<http://www.classltd.com>

Q: Which material has the highest sound speed?

Young's Modulus E 弹性模量

- E (unit: Pa): stress σ divided by strain ε
 - ▣ stress σ : force per unit area
 - ▣ strain ε : ratio of elongation



$$F = K \cdot u$$

$$\begin{aligned} E &= \frac{\sigma}{\varepsilon} = \frac{F / a^2}{u / a} = K / a \\ &= v_g^2 \frac{m}{a^3} \\ &= v_g^2 \rho \end{aligned}$$

or

$$v_g = \sqrt{\frac{E}{\rho}}$$

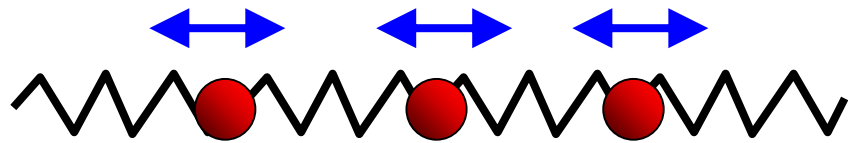
macroscopic
properties

$$v_g = a \sqrt{\frac{K}{m}}$$

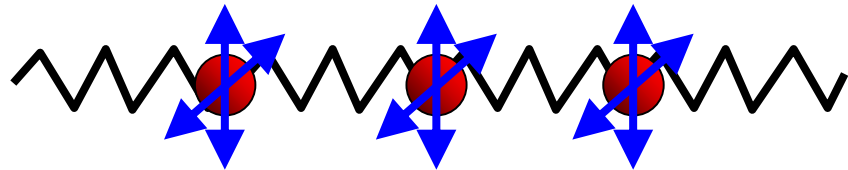
microscopic
properties

ρ - material density (kg/m³)

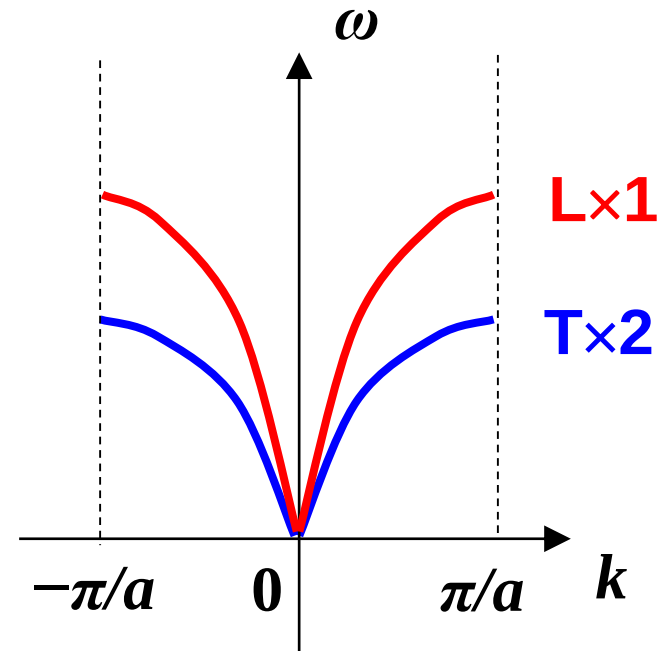
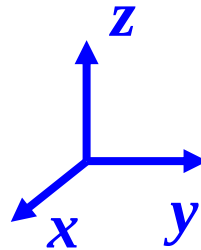
1D *Monatomic* Chain 单原子链



Longitudinal (L) 纵波 × 1



Transverse (T) 横波 × 2

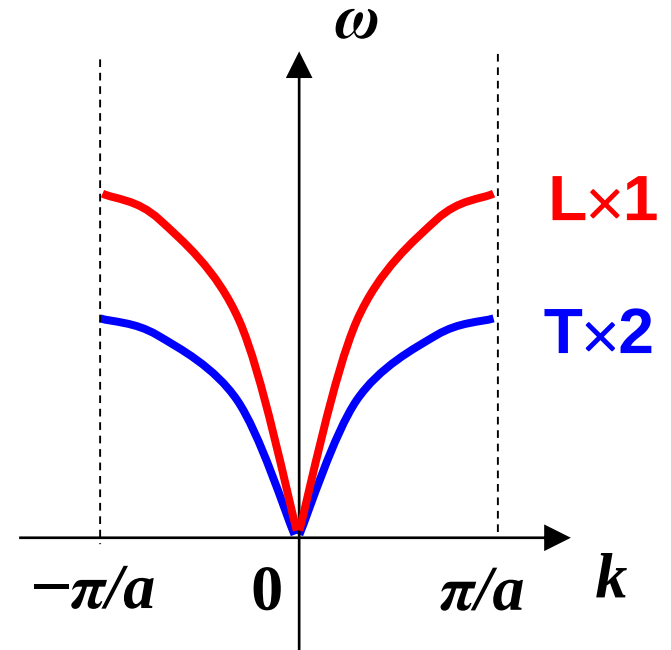
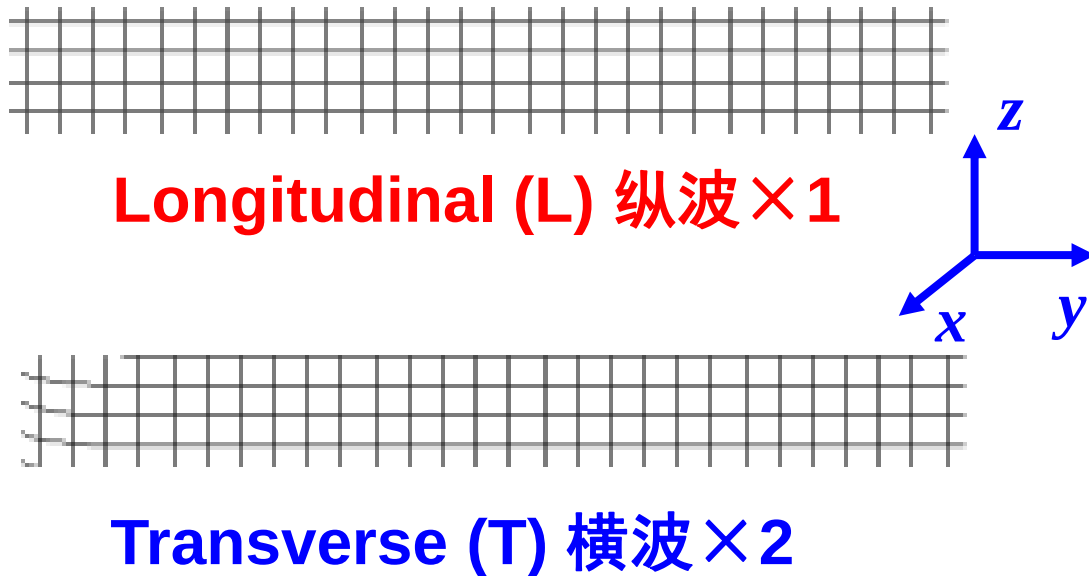


$$\omega = \sqrt{\frac{4K}{m}} \left| \sin\left(\frac{ak}{2}\right) \right|$$

$$K_L > K_T$$

$$v_{gL} > v_{gT}$$

1D *Monatomic* Chain 单原子链



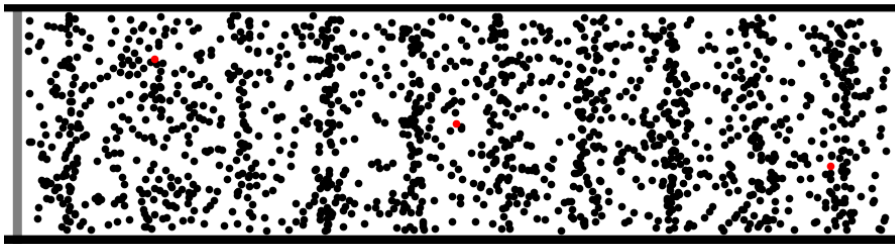
$$\omega = \sqrt{\frac{4K}{m}} \left| \sin\left(\frac{ak}{2}\right) \right|$$

$$K_L > K_T$$

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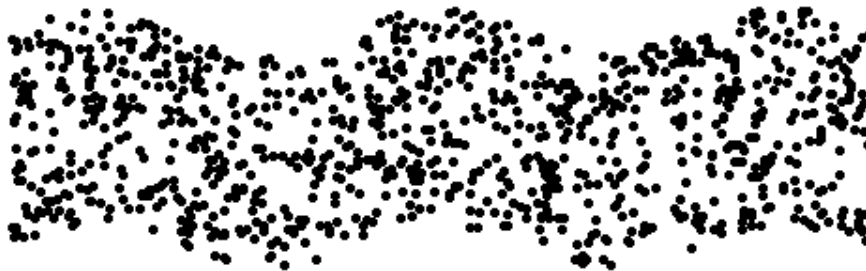
Seismic Waves 地震波

$$V_{gL} > V_{gT}$$

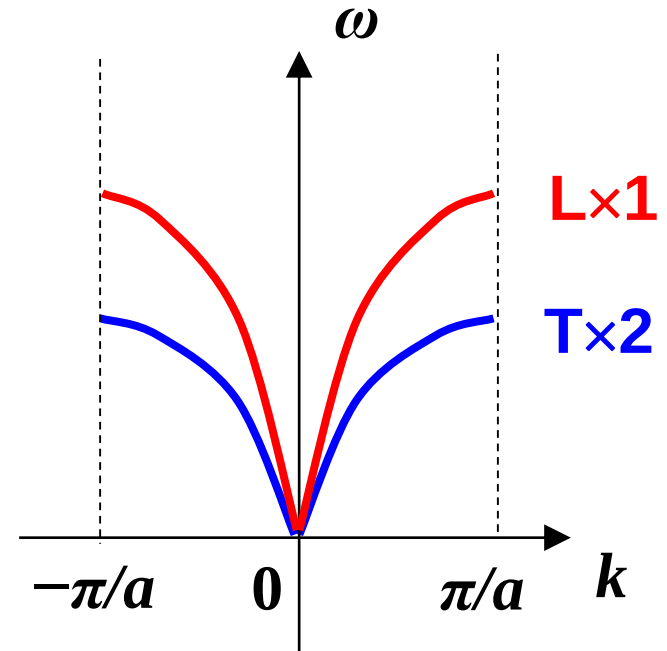


©2011, Dan Russell

L wave arrives first, less damage

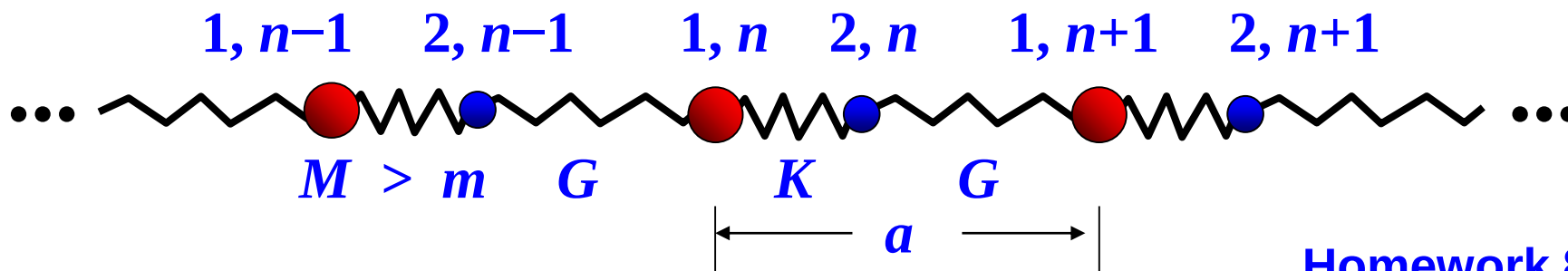


T wave arrives later, more damage!



[video - earthquake](#)

1D *Diatomic* Chain 双原子链



Homework 8.3

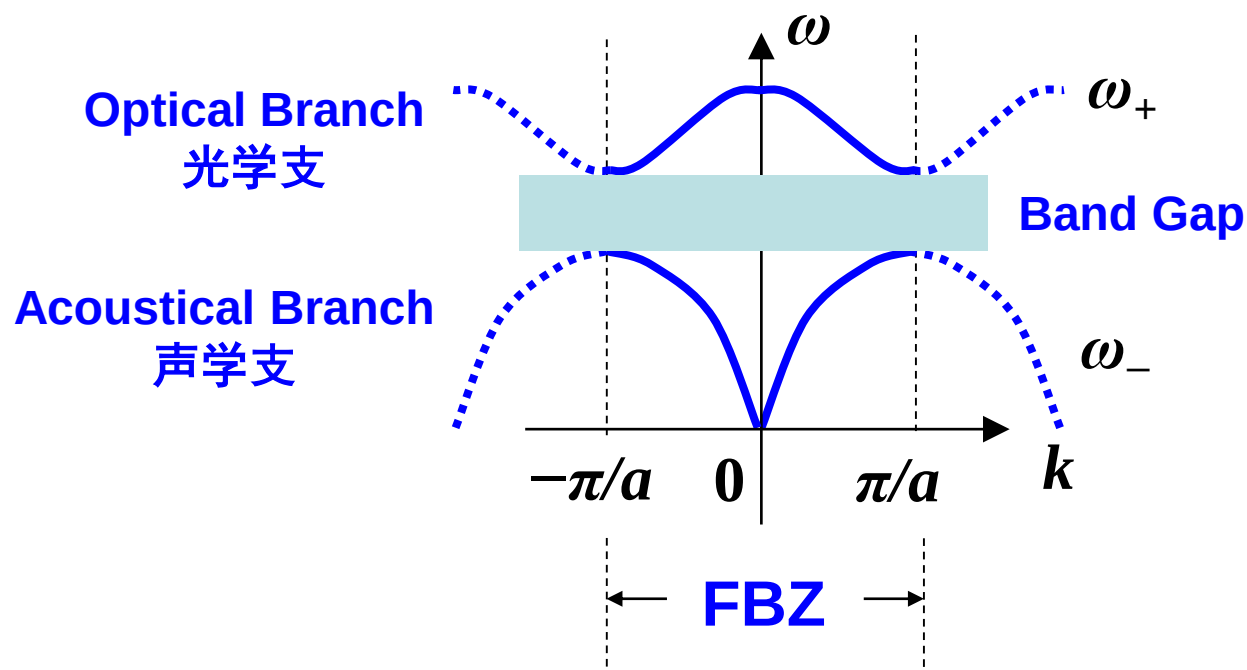
$$\left\{ \begin{array}{l} M \frac{\partial^2 u_{1,n}}{\partial t^2} + K(u_{1,n} - u_{2,n}) + G(u_{1,n} - u_{2,n-1}) = 0 \\ m \frac{\partial^2 u_{2,n}}{\partial t^2} + K(u_{2,n} - u_{1,n}) + G(u_{2,n} - u_{1,n+1}) = 0 \end{array} \right. \quad \left\{ \begin{array}{l} u_{1,n} = A_1 e^{i(kx - \omega t)} \\ u_{2,n} = A_2 e^{i(kx - \omega t)} \end{array} \right.$$

→

$$\omega^2 = \frac{(K + G)(M + m) \pm \sqrt{(K + G)^2 (M + m)^2 - 16MmKG \sin^2 \left(\frac{ak}{2} \right)}}{2Mm}$$

1D *Diatomic* Chain 双原子链

$$\omega^2 = \frac{(K + G)(M + m) \pm \sqrt{(K + G)^2(M + m)^2 - 16MmKG \sin^2\left(\frac{ak}{2}\right)}}{2Mm}$$



1D *Diatom* Chain 双原子链

When $K = G$ (example: NaCl, GaAs, ...),
Simplified solution:

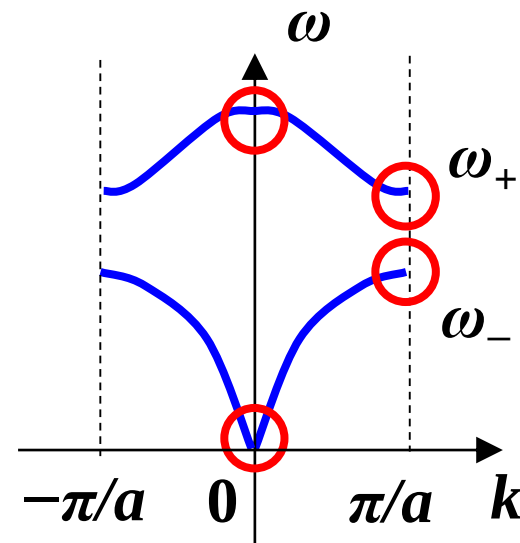
$$\omega^2 = K \frac{M+m}{Mm} \left[1 \pm \sqrt{1 - \frac{4Mm}{(M+m)^2} \sin^2 \left(\frac{ak}{2} \right)} \right]$$

$$\omega_+(k=0) = \sqrt{2K \frac{M+m}{Mm}}$$

$$\omega_+(k = \pm \frac{\pi}{a}) = \sqrt{\frac{2K}{m}}$$

$$\omega_-(k \approx 0) = ak \sqrt{\frac{K}{2(M+m)}}$$

$$\omega_-(k = \pm \frac{\pi}{a}) = \sqrt{\frac{2K}{M}}$$



**Note: only the primitive cell matters;
the distance between atoms 1 and 2 does not affect results**

1D *Diatom* Chain 双原子链

When $K = G$ (example: NaCl, GaAs, ...),
Simplified solution:

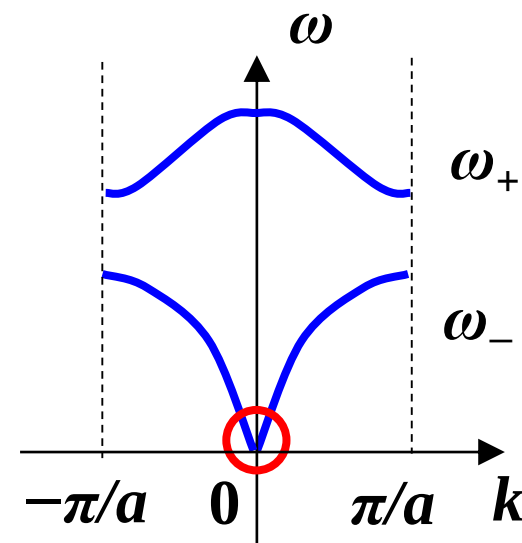
$$\omega^2 = K \frac{M+m}{Mm} \left[1 \pm \sqrt{1 - \frac{4Mm}{(M+m)^2} \sin^2 \left(\frac{ak}{2} \right)} \right]$$

$$\omega_-(k \approx 0) = ak \sqrt{\frac{K}{2(M+m)}}$$

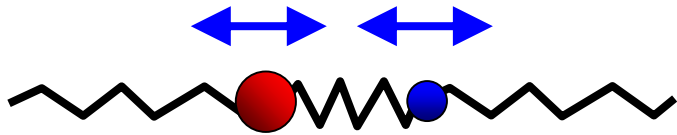
Sound Speed

$$v_g = \partial \omega / \partial k$$

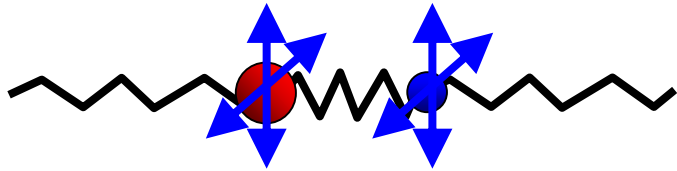
$$= a \sqrt{\frac{K}{2(M+m)}}$$



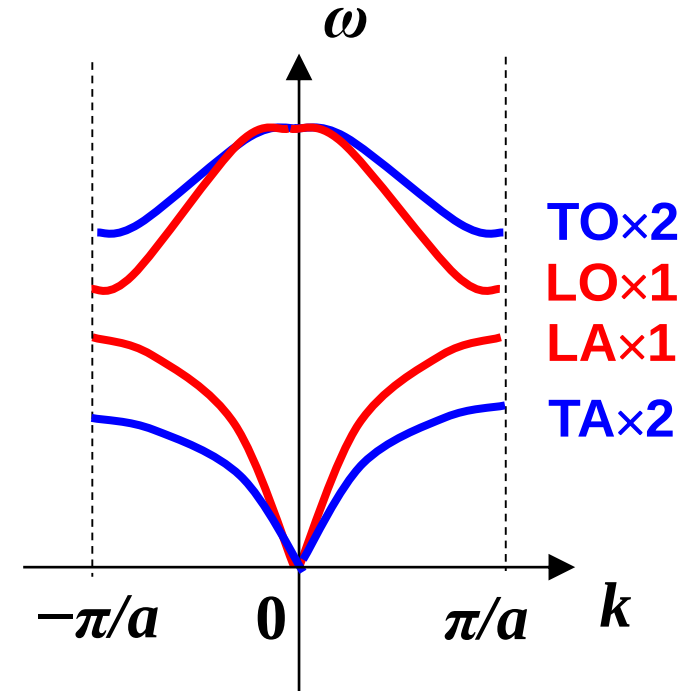
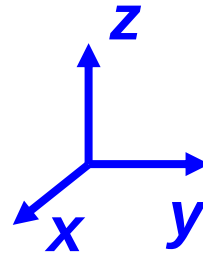
1D *Diatomic* Chain 双原子链



Longitudinal (L) 纵波 $\times 1$

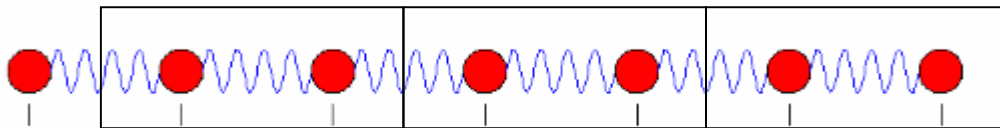


Transverse (T) 横波 $\times 2$

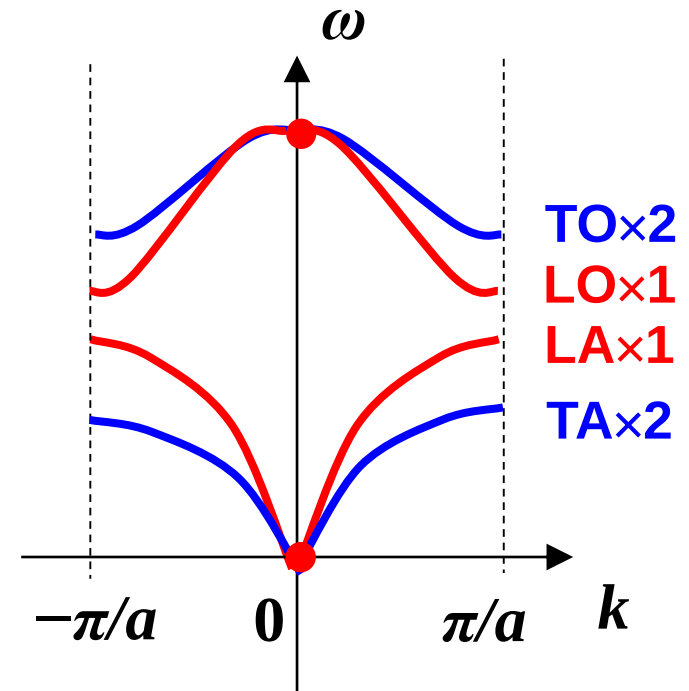
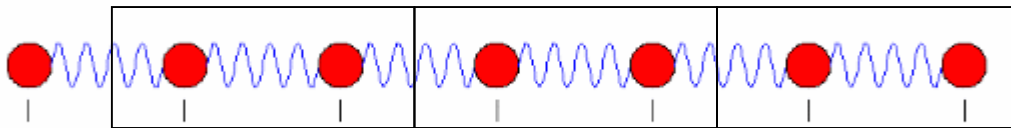


1D *Diatomic* Chain 双原子链

Longitudinal Optical (LO)



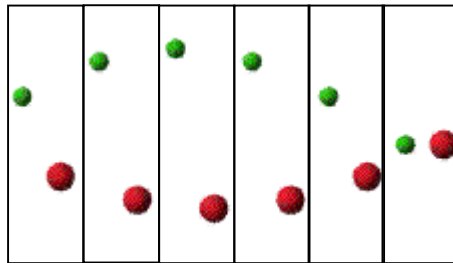
Longitudinal Acoustical (LA)



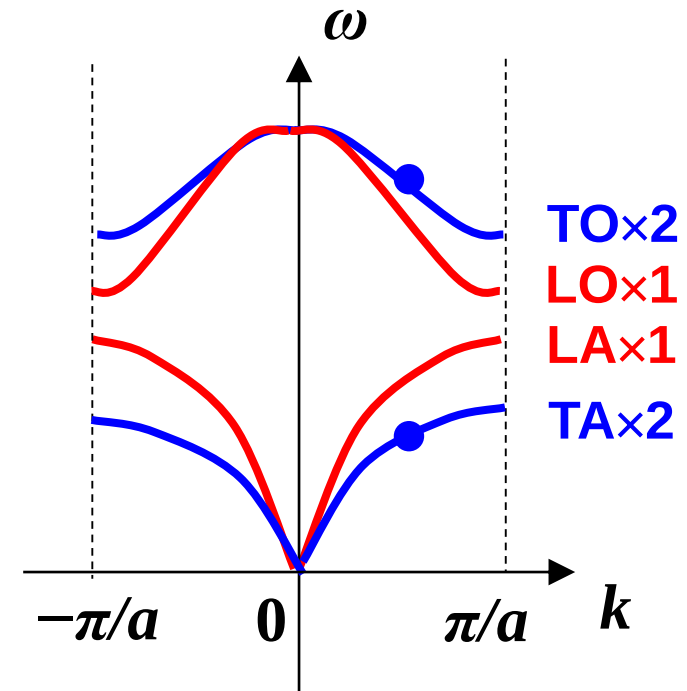
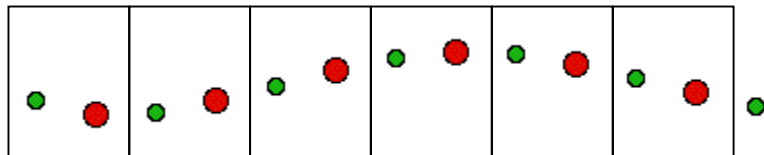
Acoustic modes are related to the low frequency vibration across the entire crystal;
Optical modes are related to the high frequency vibration inside the primitive cell.

1D *Diatomic* Chain 双原子链

Transverse Optical (TO)

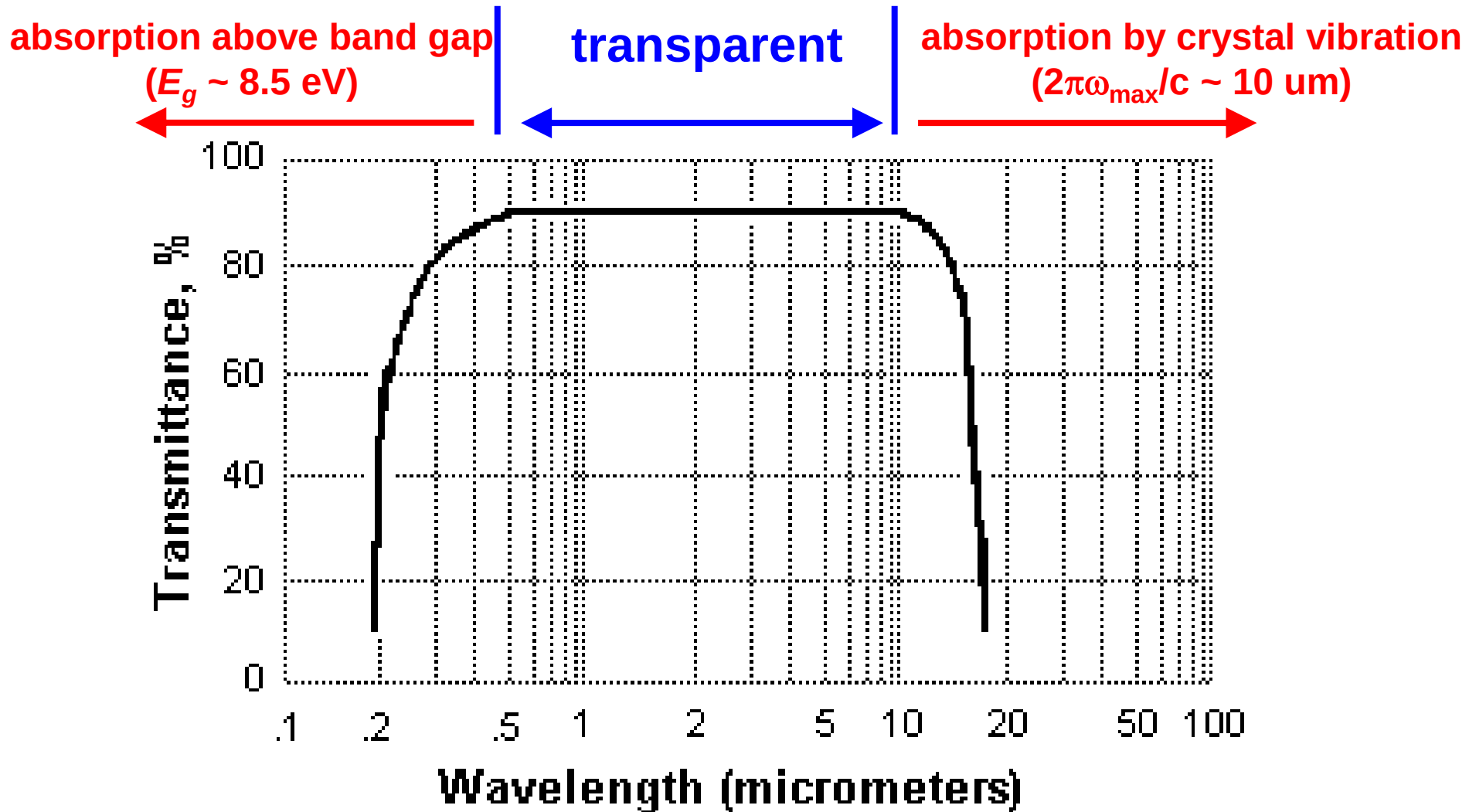


Transverse Acoustical (TA)

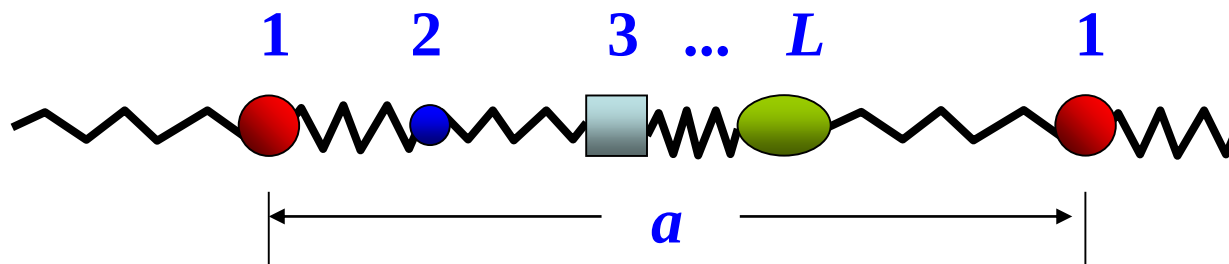


Acoustic modes are related to the low frequency vibration across the entire crystal;
Optical modes are related to the high frequency vibration inside the primitive cell.

Optical Properties of NaCl



1D *Multi-atomic* Chain 多原子链

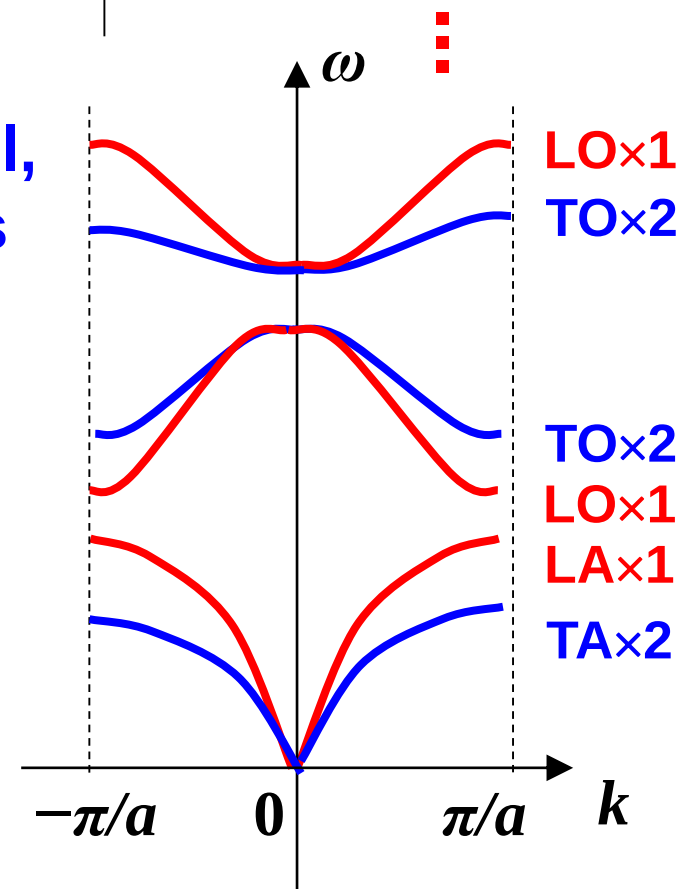


If there are L atoms in a primitive cell,
there will be more optical branches
(but still finite)

There will be:

1 LA, 2 TA

$(L-1)$ LO, $2(L-1)$ TO



Fundamentals of Solid State Physics

Lattice Vibration - Quantum Model

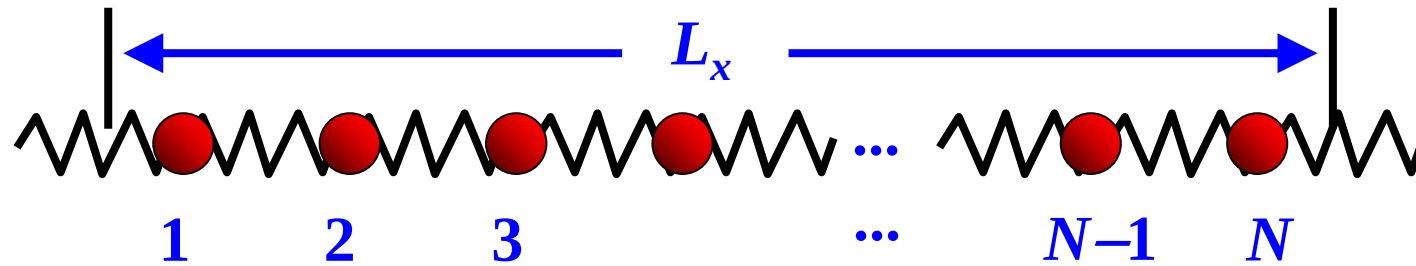
Xing Sheng 盛兴



Department of Electronic Engineering
Tsinghua University

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Quantization of the Bands 量子化



$$L_x = N \cdot a, N \text{ is large } \sim 10^{23}$$

Born-von Karman *periodic* boundary condition

$$u(x) = u(x + L_x)$$



$$\exp(ik_x L_x) = 1$$

$$u(x) = Ae^{i(kx - \omega t)}$$



$$k_x = \frac{2\pi n_x}{L_x}$$

$$n_x = 0, \pm 1, \pm 2, \dots$$

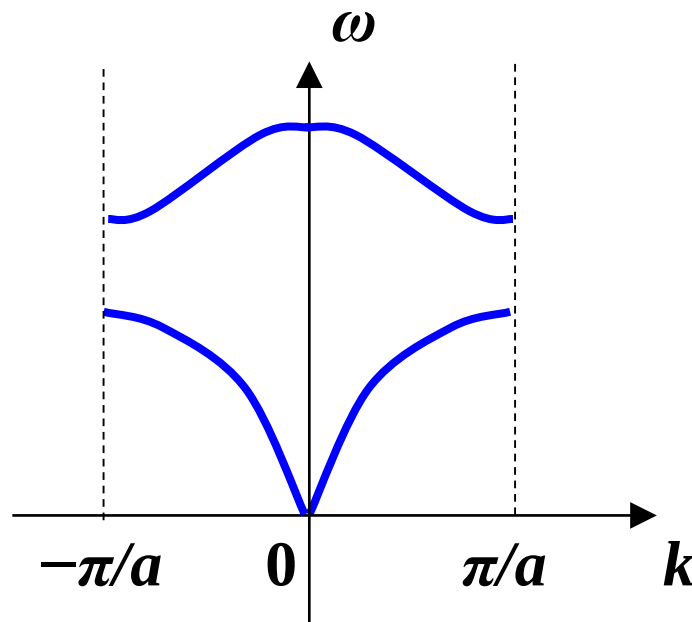
k is a *quantized* value

Quantization of the Bands 量子化

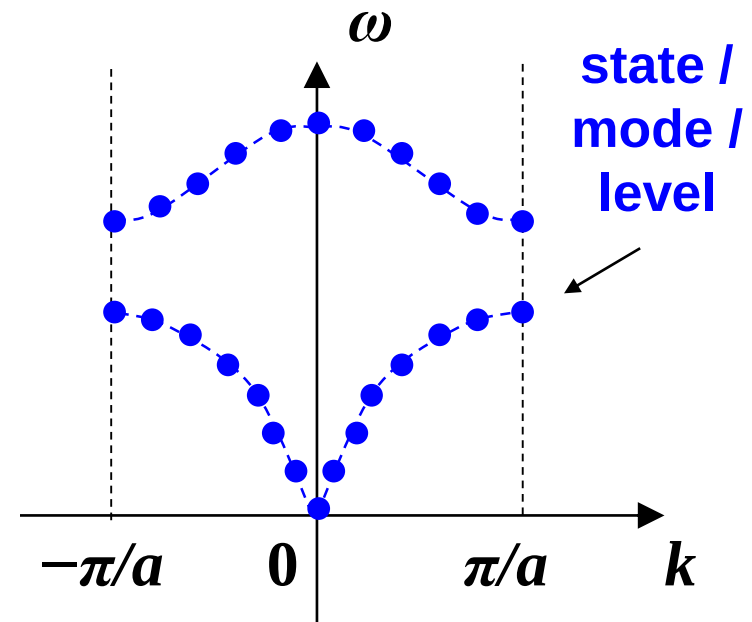
$$k_x = \frac{2\pi n_x}{L_x}$$

$$n_x = 0, \pm 1, \pm 2, \dots$$

quasi-continuous



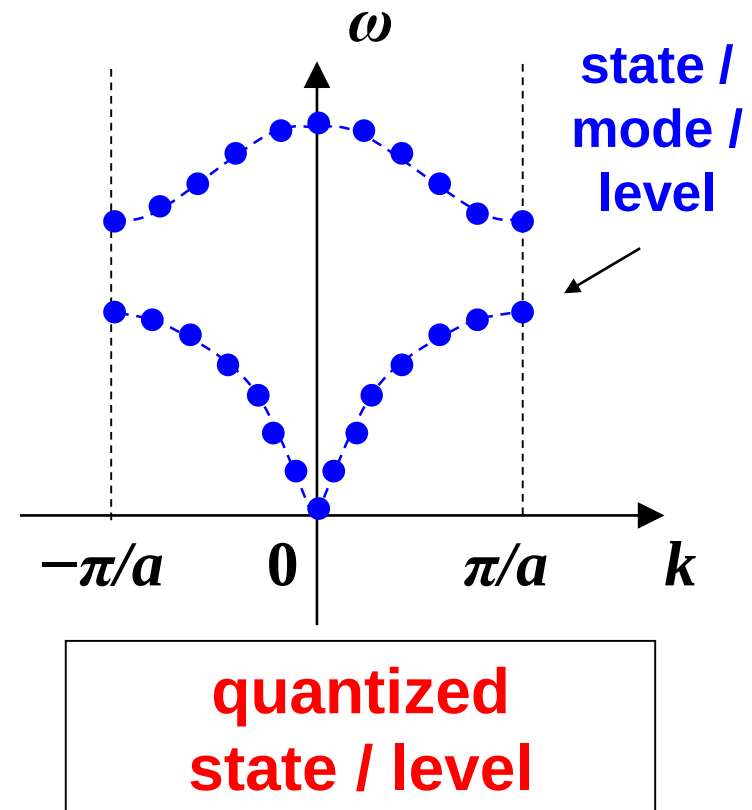
ω - k diagram
(dispersion curve)



**quantized
state / level**

Phonon Band Diagram

- A vibration state (phonon) is a collective movement of all the atoms in the lattice, not vibrations from a single atom
- There are N states in each band (N : number of primitive cells in the crystal)
- If there are L atoms in each primitive cell, there will be **$3L$ bands, and $3NL$ states**
 - 1 LA + 2 TA
 - $(L-1)$ LO + $2(L-1)$ TO



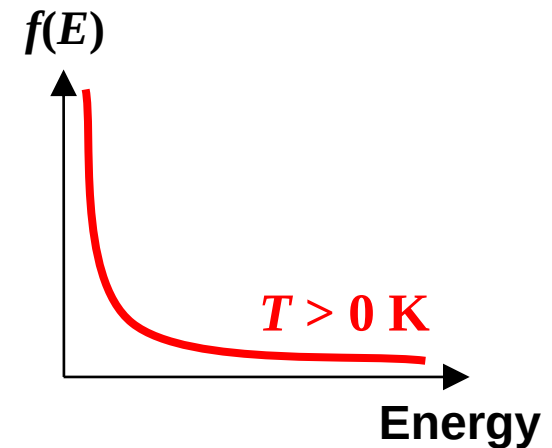
Phonon 声子

- Phonon is a boson

- Each state can fill many phonons at the same time
- Bose-Einstein Distribution

number of
phonons

$$f(E = \hbar\omega) = \frac{1}{e^{\hbar\omega/k_B T} - 1}$$



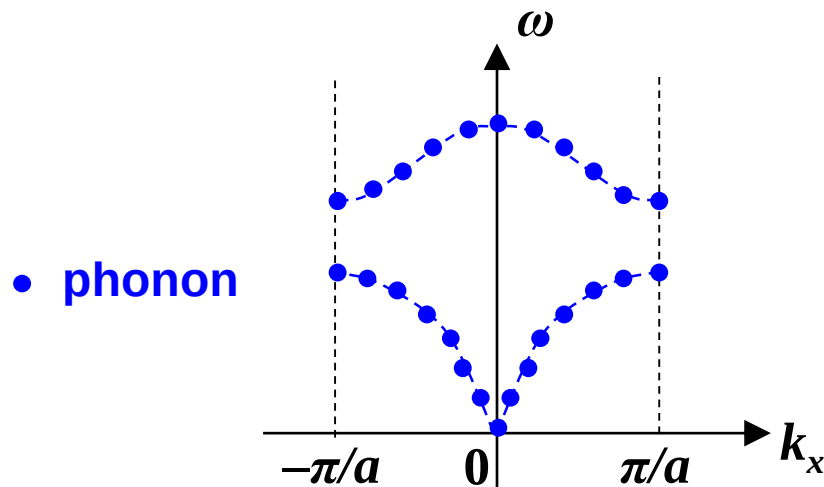
- Phonon 声子 vs. Photon 光子

- Both of them are bosons, but
- Phonon is a collective atom vibration, quasi-particle (准粒子)
- Photon is a fundamental particle (基本粒子)

Phonon Band vs. Electron Band

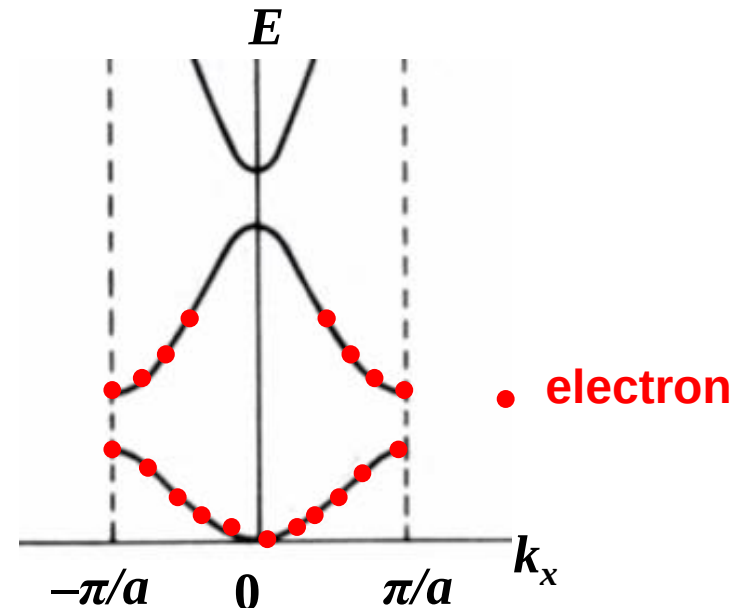
■ Phonon Band

- ❑ max frequency $\omega_{\max}$
- ❑ phonon number is not constant, depend on T
- ❑ each state can have many phonons (Bosons)



■ Electron Band

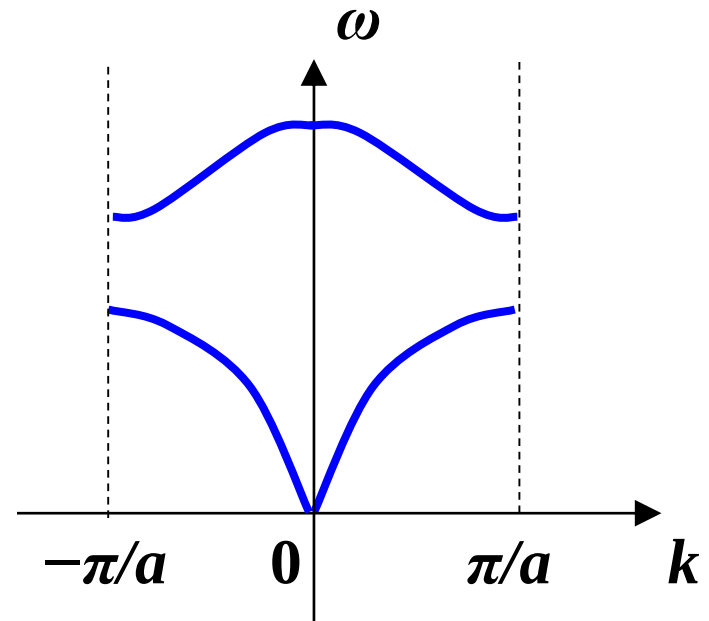
- ❑ no highest energy
- ❑ electron number is fixed
- ❑ each state can only have one or zero electrons



k values have the same physical meaning, related to the crystal structure

Measure Phonon Band Diagram

- Optical Scattering
 - ▣ Brillouin Scattering 布里渊散射
 - ▣ Raman Scattering 拉曼散射
- Neutron Scattering 中子散射



Measure Phonon Band Diagram

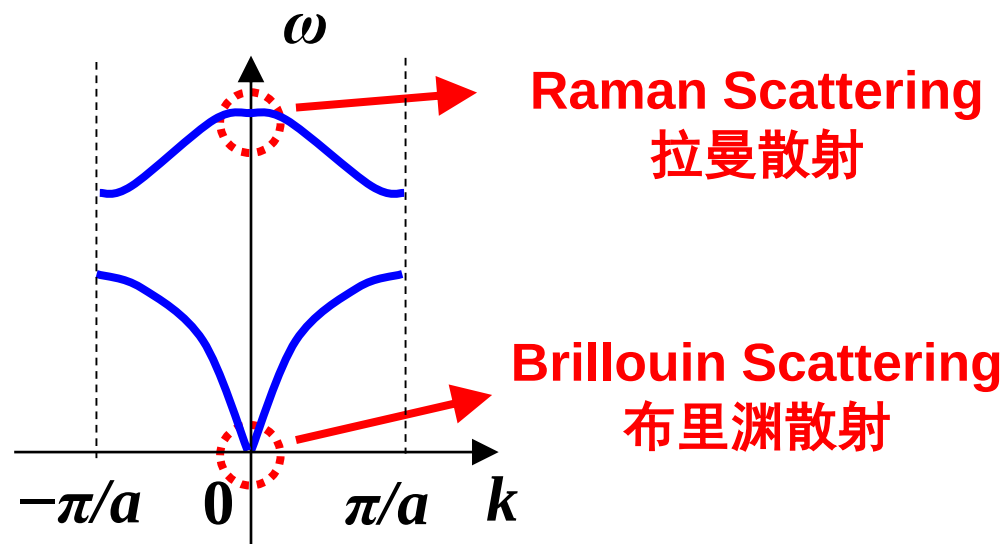
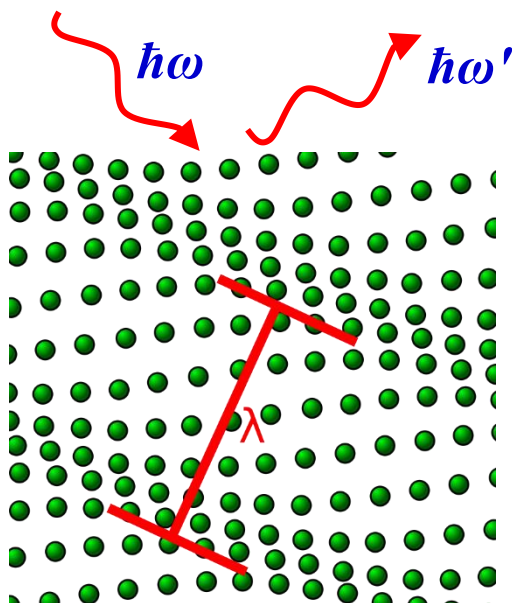
■ Optical Scattering

- ❑ Photons have much smaller momentum than phonons
- ❑ can only measure $k \approx 0$

$$p = \frac{h}{\lambda} \ll \frac{h}{a}$$

$$\lambda \gg a$$

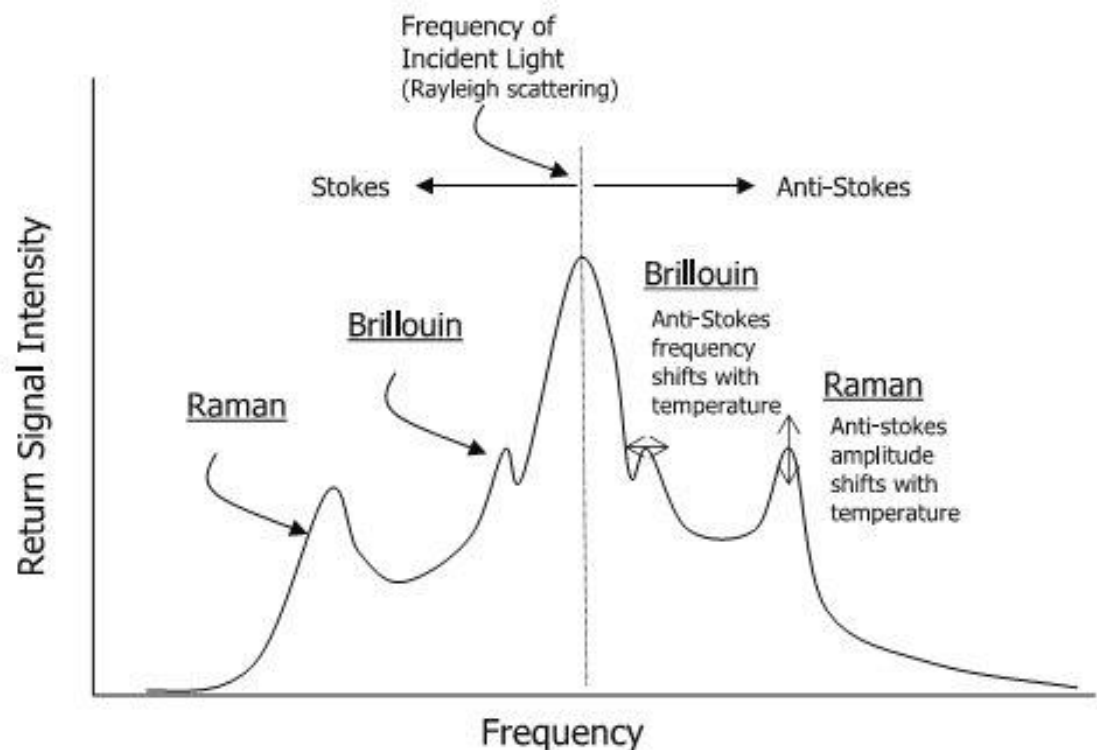
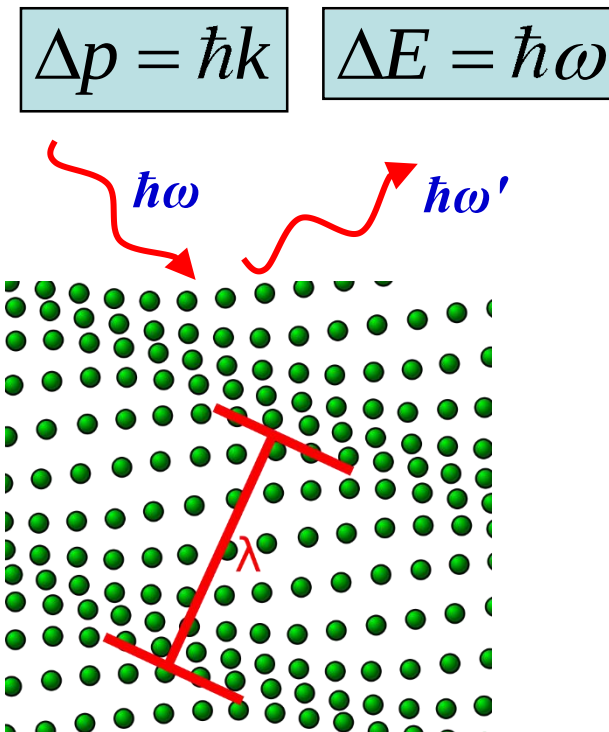
$$\Delta p = \hbar k \quad \Delta E = \hbar \omega$$



Measure Phonon Band Diagram

■ Optical Scattering

- ❑ Photons have much smaller momentum than phonons
- ❑ can only measure $k \approx 0$



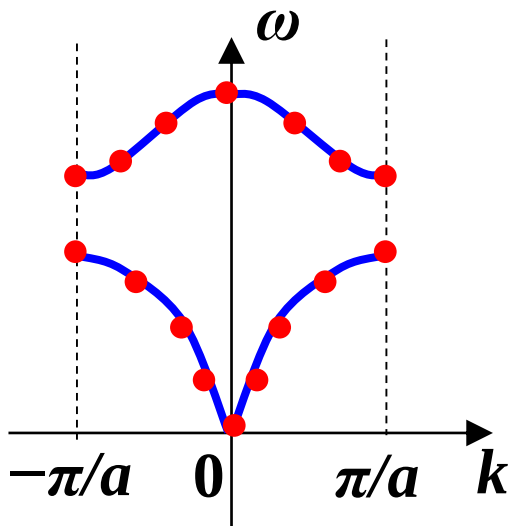
Neutron Scattering 中子散射

- Neutron has a similar mass with atoms
- Momentum and energy can cover the entire bands

$$\Delta p = \hbar k$$

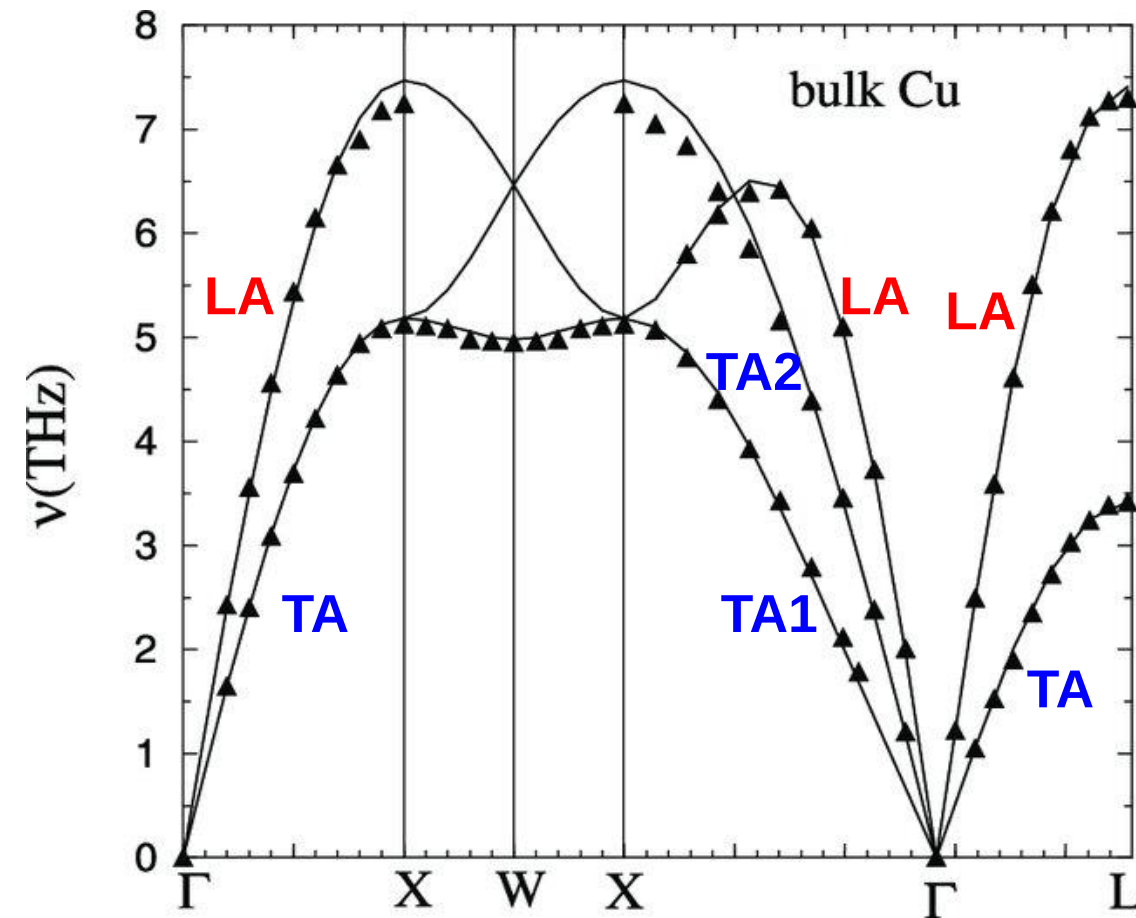
$$\Delta E = \hbar \omega$$

- Need a nuclear reactor!



散裂中子源，东莞

Phonon Band Diagram - Copper

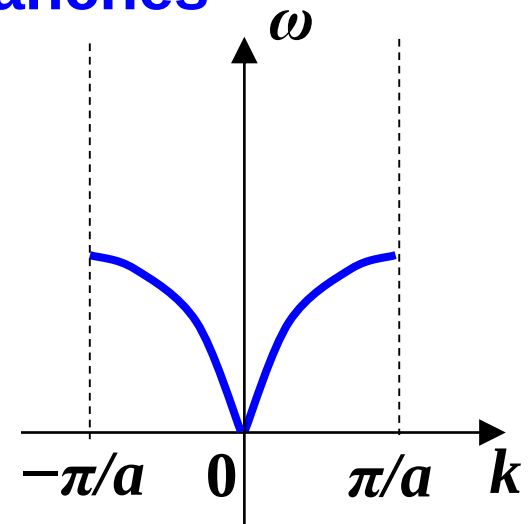


▲ measured by neutron scattering

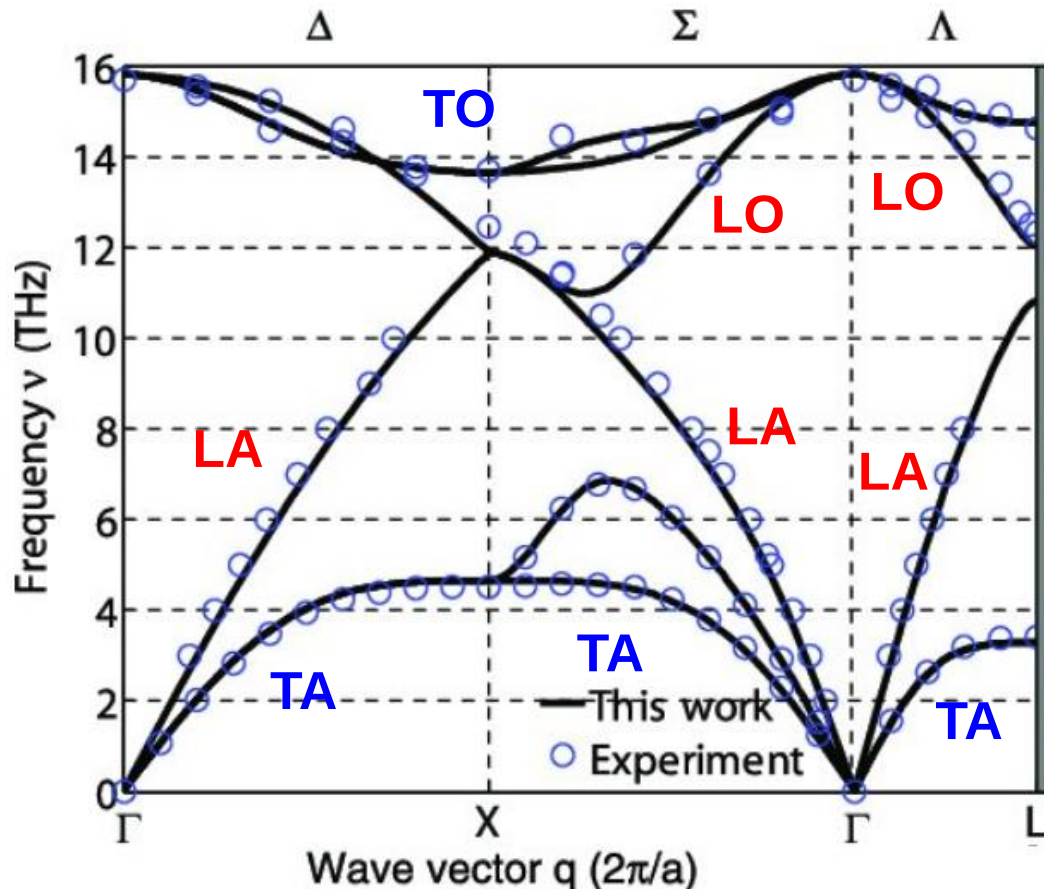
— calculation

Cu has FCC structure, with **one** atom in a primitive cell.

There are only acoustic branches, no optical branches

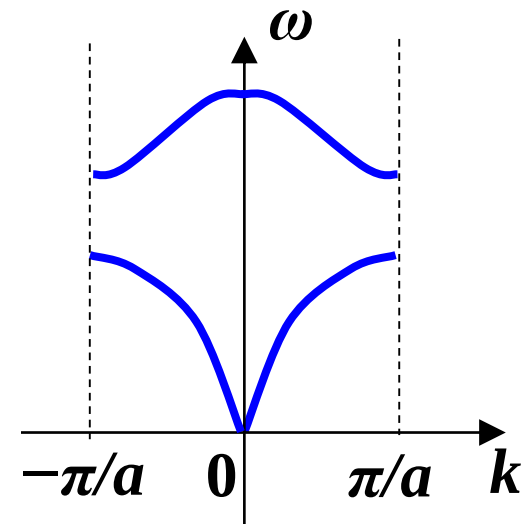


Phonon Band Diagram - Silicon



Si has FCC structure, but with **two** atoms in a primitive cell.

There are both acoustic and optical branches.



○ measured by neutron scattering
— calculation

Phonon Density of States (DOS)

$$g(\omega) = \frac{dn}{d\omega}$$

DOS - number of states/levels per unit frequency in $[\omega, \omega + d\omega]$, per unit volume

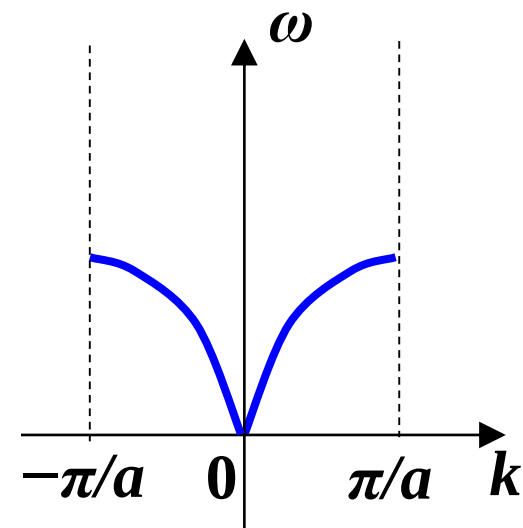
**For 1D chain
LA mode**

$$\omega = \sqrt{\frac{4K}{m}} \left| \sin\left(\frac{ka}{2}\right) \right|$$

$$n = \frac{N}{L_x} = \frac{2k}{2\pi / L_x} \cdot \frac{1}{L_x} = \frac{k}{\pi}$$

→

$$g(\omega) = \frac{dn}{d\omega} = \frac{dn}{dk} \bigg/ \frac{d\omega}{dk} = \dots$$



Phonon Density of States (DOS)

$$g(\omega) = \frac{dn}{d\omega}$$

DOS - number of states/levels per unit frequency in $[\omega, \omega + d\omega]$, per unit volume

For 3D solid

$$\omega = \sqrt{\frac{4K}{m}} \left| \sin\left(\frac{ka}{2}\right) \right|$$

$$n = \frac{3N}{V} = \frac{\frac{4\pi}{3}k^3}{\frac{(2\pi)^3}{V}} = \frac{1}{2\pi^2}k^3$$

**at low ω
limit**



$$\omega \approx v_g k = v_g (2\pi^2 n)^{1/3}$$

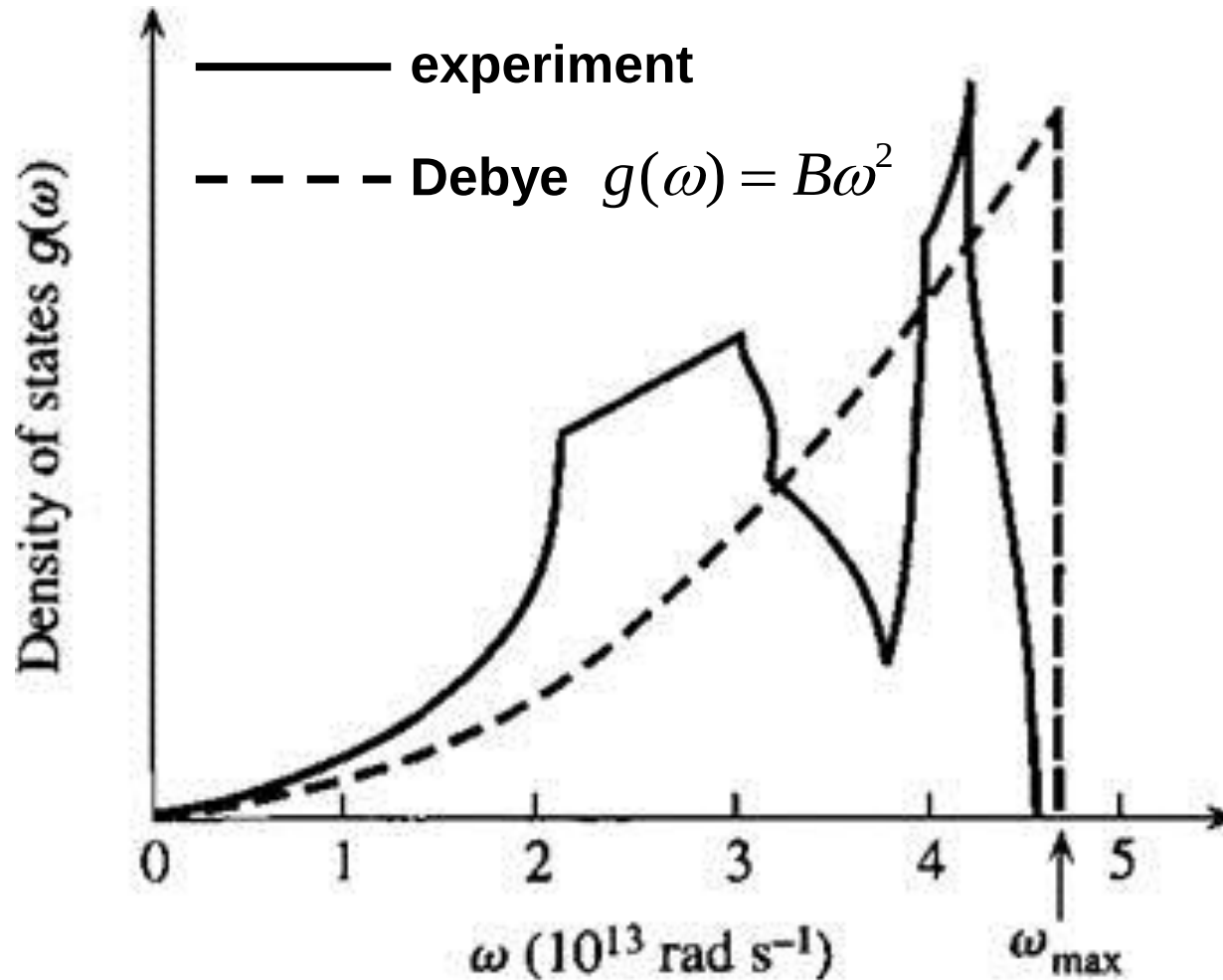


$$g(\omega) = \frac{dn}{d\omega} = \frac{3}{2\pi^2 v_g^3} \omega^2 = B\omega^2$$

**The Debye Model
(德拜模型)**

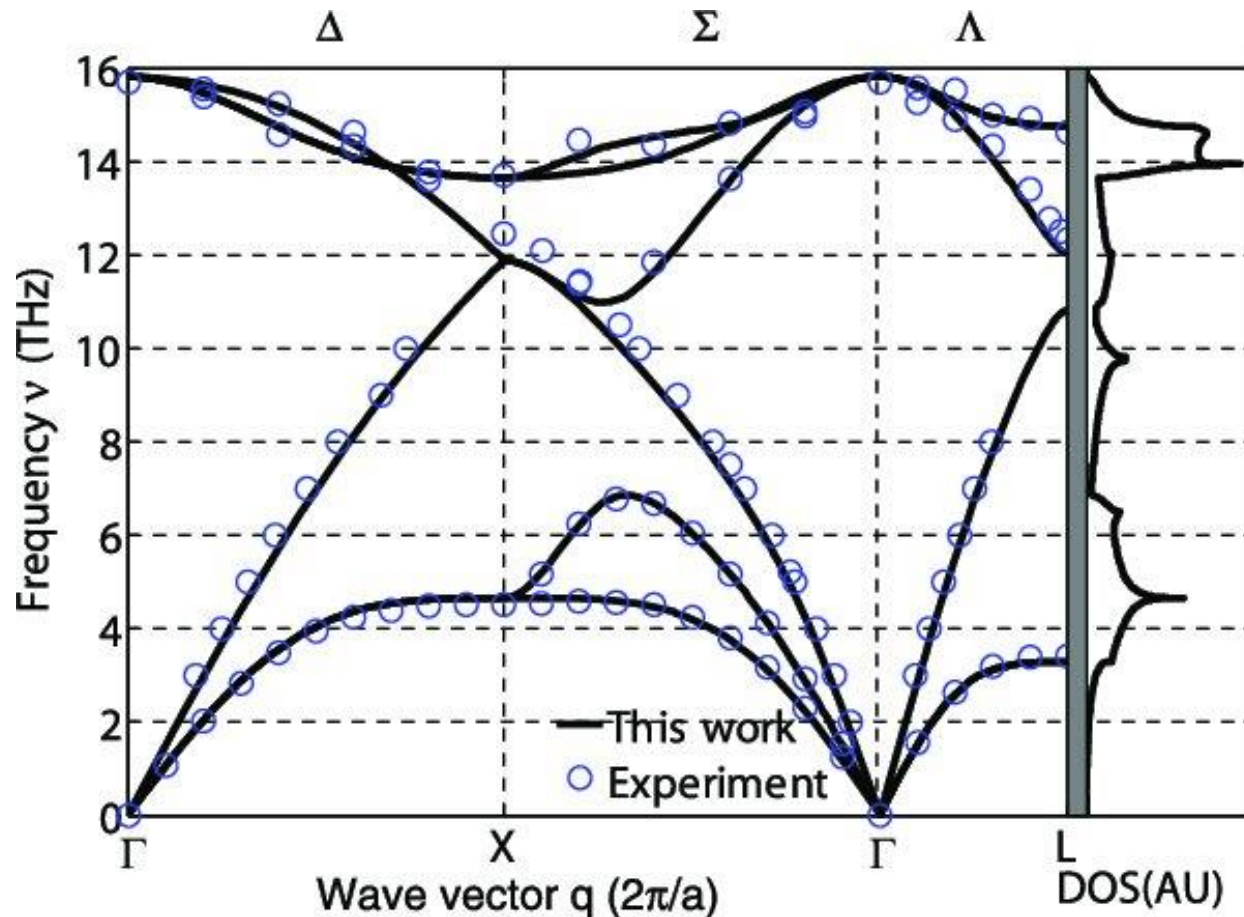
Phonon Density of States (DOS)

DOS for copper



Phonon Density of States (DOS)

Phonon band diagram and DOS for Silicon



Thank you for your attention